Electrochemistry

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Electrochemistry

Introduction

Electrochemistry = (Electrical Energy) + (Chemical Energy)

The study of interconversion between electrical energy and chemical energy.

The devices used for this interconversion are called electrochemical cells which are of following two types :

	Galvanic cell	Electrolytic cell
(i)	Converts chemical energy into electrical energy.	Converts electrical energy into chemical energy.
(ii)	Spontaneous redox reaction takes place.	Non-spontaneous redox reaction takes place.
(iii)	Gibbs free energy of reaction at constant temperature and pressure $(\Delta_r G)_{T,P}$ $(\Delta_r G)_{T,P} < 0$ (Negative)	$(\Delta_r G)_{T,P} > 0$
(iv)	Anode → (−)ve Cathode → (+)ve	Anode → (+)ve Cathode → (−)ve

Galvanic Cell :

A galvanic cell is made by joining two half-cells.

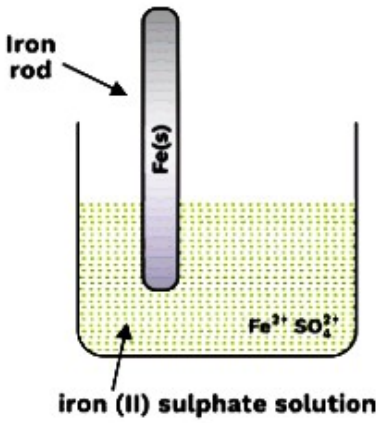
Half-cell of a substance consists an electrode of that substance dipped in aqueous Solution containing ions of same substance.

Points to remember

In any electrochemical call
At anode → always oxidation
At cathode → always reduction

Example :

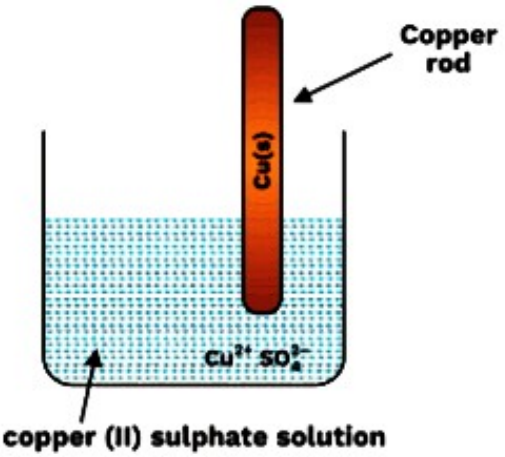
Iron half-cell
Iron rod dipped in $FeSO_4(aq)$ solution
(Fe^{+2}) cannot exist independently.



Example :

Cu half-cell

Copper rod/wire dipped in $Cu^{+2}(aq)$ (from $CuSO_4$) solution



In a half-cell, there is a natural tendency of substance to lose or gain electrons at the surface of electrode.

Interface of rod and aq. solution is called surface

Iron generally, has tendency to loose electron

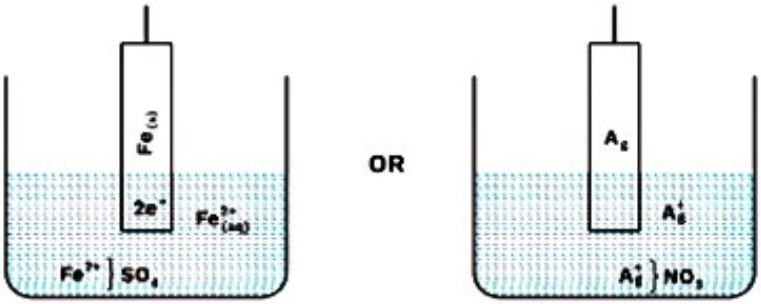
$$Fe_{(s)} \rightarrow Fe^{+2}_{(aq.)} + 2e^-$$

rod

Silver has natural tendency to gain electron

$$Ag^+ + e^- \rightarrow Ag(s)$$

(from $AgNO_3$) deposit on surface



Initially Fe(s) neutral
$FeSO_4$ solution (neutral)

Later

$$Fe(s) \underset{\text{(rod) (negative)}}{\longrightarrow} \underset{\text{(aq) (positive) surface}}{Fe^{+2}} + \underset{\text{Potential differential generated at surface}}{2e^-}$$

Initially Ag(s) neutral
$AgNO_3$ solution-neutral

$$\underset{\text{(aq) (negative) at surface}}{Ag^+} + e^- \longrightarrow \underset{\text{rod (positive)}}{Ag(s)}$$

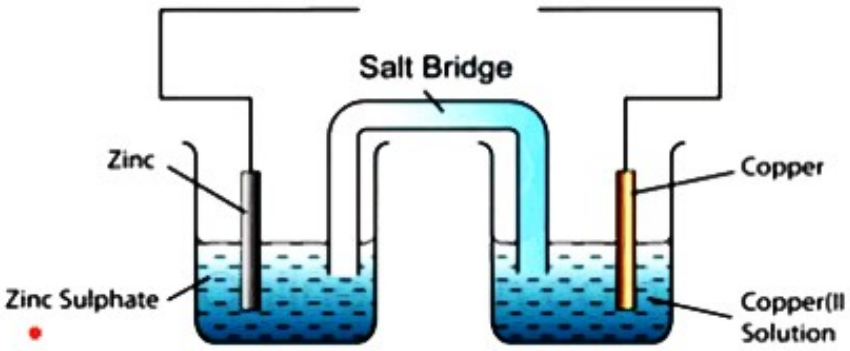
In a half-cell, a potential difference is developed between electrode and the solution which is called electrode potential. When two half cells are connected then a galvanic cell is formed. Various combinations of cells can be created. Some are discussed in detail below.

Daniell Cell (Zn-Cu cell):

Wire – external circuit

Zinc : natural tendency to loose electrons

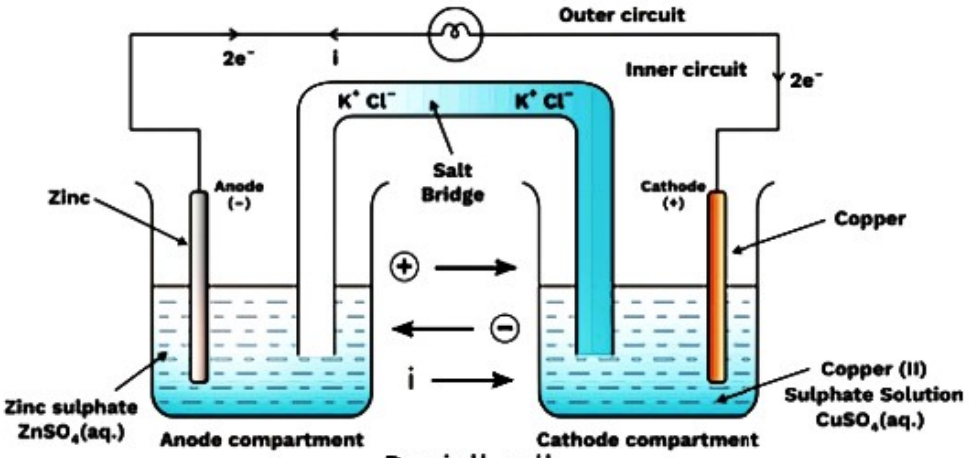
$$\underset{\text{(Negative)}}{Zn(s)} \longrightarrow \underset{\text{(aq)}}{Zn^{+2}} + 2e^-$$



Copper : natural tendency to gain e^-

$$Cu^{+2} + 2e^- \longrightarrow \underset{\text{Positive}}{Cu(s)}$$

On connecting opposite polarity electrode, flow of current takes place
Negative polarity = anode
Positive polarity = cathode



Daniell cell

Working :

On connecting electrons from Zn rod reaches Cu^{+2} ($CuSO_4$)

$Cu^{+2} + 2e^- \longrightarrow Cu(s)$ deposits on copper rod

flow of electrons in external circuit (Zn rod to Cu)

Salt Bridge

For further flow of current, electrical neutrality required in both half-cell. Salt bridge is used to accomplish that. It is a glass tube, U shaped, filled with strong electrolyte in semi solid paste of strong electrolytes like KCl and KNO_3 etc.

Example : KCl

For further flow of current, electrical neutrality required in both half cell.

SO_4^{-2} balanced by $2K^+$ from salt bridge

Zn^{+2} (aq) balanced by $2Cl^-$ from salt bridge

Working :

At anode, Zn comes out as Zn^{2+} into the solution leaving 2 electrons at the surface of electrode.

These two electrons travel through external circuit, reach at cathode, where Cu^{2+} of solution captures them and gets deposited on the surface of cathode.

To maintain electrical neutrality in both compartment, ions from salt bridge enter into solutions due to which the cell reaction keeps running.

After some time, when static or dynamic equilibrium achieved, flow of current stops.

Anode : $Zn(s) \rightleftharpoons Zn^{2+}_{(aq.)} + 2e^-$ (Oxidation)

Cathode : $Cu^{2+}_{(aq.)} + 2e^- \rightleftharpoons Cu(s)$ (Reduction)

Net cell reaction (NCR): $Zn(s) + Cu^{2+}_{(aq.)} \rightleftharpoons Zn^{2+}_{(aq.)} + Cu(s)$ (Redox)

Cell representation : $Zn(s) \mid Zn^{2+}_{(aq.)} \parallel Cu^{2+}_{(aq.)} \mid Cu(s)$

Anode Cathode

Uses of salt bridge :

(i) It completes the circuit of the cell.

(ii) It maintains electrical neutrality in both compartment by movement of its ions in opposite direction.

Example :

Anode compartment contains excess of (+) ve charge ⇒ anions from salt-bridge enter into anode compartment to neutralize those charges.

Required properties by electrolyte of salt bridge :

(i) Speeds of both type ions furnished by electrolyte, should be nearly same.

(ii) Ions of electrolyte must be chemically inert with substances of compartment.

(iii) Ions of electrolyte should not show any electrochemical change with substances of compartment.

(iv) Ions of salt bridge should not participate in oxidation and reduction, if they do the potential calculated will be incorrect.

Points to remember

IUPAC representation of cell

Anode || Cathode

Daniel cell

$Zn(s) \mid Zn^{2+}_{(aq)} \parallel Cu^{2+}_{(aq)} \mid Cu(s)$

- anode ⇒ oxidation (i.e., O.N. ↑)
- cathode ⇒ reduction (i.e., O.N. ↓)

Hence, only some type of electrolyte can be used as salt bridge.

$Zn(s) \longrightarrow \underset{(aq)}{Zn^{+2}} + 2e^-$ } Cl^- from salt bridge donot form $ZnCl_2$

Q1 **In a galvanic cell, the salt bridge**
(A) does not participate chemically in the cell reaction.
(B) stops the diffusion of ions from one electrode to another.
(C) is necessary for the occurrence of the cell reaction.
(D) ensures mixing of the two electrolytic solutions

Sol. **(AC)**

Salt bridge does not participate in a chemical reaction.

Does not stop diffusion of ions.

Necessary else both compartments will have excess of charge preventing from of current. No mixing of solutions.

Electrode Potential (E) :

The natural tendency of a substance to loose or gain e^- in its half cell.
Measured in volts.
An intensive property.
There are two types as following :

(i) For oxidation $\Rightarrow$ E_{OP} (oxidation potential)
E.g. $E_{Zn|\,Zn^{2+}}$ = O.P of Zn

(ii) For reduction $\Rightarrow$ E_{RP} (reduction potential)
E.g. $E_{Cu^{2+}|Cu}$ = R.P of Cu^{2+}

Standard Electrode Potential (E°)

It is electrode potential measured at standard condition.
i.e., Concentration = 1 M
P = 1 atm
T = any given temp (by default 298 K)
It is also of two types :
i.e., $E^\circ_{Zn/Zn^{2+}}$ $\Rightarrow$ SOP (Std. oxidation potential of Zn)

$E^\circ_{Cu^{2+}/Cu}$ $\Rightarrow$ SRP (Std. reduction potential of Cu^{2+})

For a substance pair,

$$\boxed{E^\circ_{OP} = (-E^\circ_{RP})}$$

i.e., $E^{\circ}_{Zn/Zn^{2+}} = -E^{\circ}_{Zn^{2+}/Zn}$

$E^{\circ}_{Cu^{2+}/Cu} = -E^{\circ}_{Cu/Cu^{2+}}$

At a certain temperature, each substance has certain fixed value of its E° which depends upon temperature only.

Example :

$E^{\circ}_{Zn/Zn^{2+}} = 0.76 \quad \Rightarrow E^{\circ}_{Zn^{2+}/Zn} = -0.76\,V$

$E^{\circ}_{Fe/Fe^{2+}} = 0.44\,V \quad \Rightarrow E^{\circ}_{Fe^{2+}/Fe} = -0.44\,V$

$E^{\circ}_{Cu^{2+}/Cu} = 0.34\,V$

$E^{\circ}_{Ag^{+}/Ag} = 0.8\,V \quad \Rightarrow E^{\circ}_{Ag/Ag^{+}} = -0.8\,V$

Electromotive force of cell (emf of cell) :

It is the ability of a galvanic cell to push electrons from anode to cathode through external circuit.
It is actually the difference between electrode potentially of both half calls.
Means measured in volts.

For a galvanic cell

At non-standard state $\boxed{E_{cell} = E_{OP} + E_{RP}}$

At standard state $\boxed{E^{\circ}_{cell} = E^{\circ}_{OP} + E^{\circ}_{RP}}$

Where,

E_{OP} or E°_{OP} → For substance oxidised

E_{RP} or E°_{RP} → For substance reduced

Example : For Daniel cell

$Zn(s) \mid Zn^{2+}_{(aq.)} \parallel Cu^{2+}_{(aq.)} \mid Cu(s)$

$E^{\circ}_{cell} = E^{\circ}_{OP} + E^{\circ}_{RP}$

$= \left(E^{\circ}_{Zn/Zn^{2+}}\right) + \left(E^{\circ}_{Cu^{2+}/Cu}\right)$

$= (0.76) + (0.34)$

$= 1.1$ volts

Points to remember

These values of E° are determined w.r.t. hydrogen for which, according to IUPAC convention, the value of E° at all temperatures is assigned zero.

i.e., $\boxed{E^{\circ}_{H_2/H^{+}} = 0.0\,volt = E^{\circ}_{H^{+}/H_2}}$ at all temperature

Q2 **Given :** $E^{\circ}_{Zn^{2+}/Zn} = -0.76\,V$; $E^{\circ}_{Fe^{2+}/Fe} = -0.44\,V$

$E^{\circ}_{Cu^{2+}/Cu} = 0.34\,V$; $E^{\circ}_{Ag^{+}/Ag} = 0.80\,V$

Determine E°_{cell} for following cells :

1) $Zn(s) \mid Zn^{2+}(aq) \parallel Ag^{+}(aq) \mid Ag(s)$

2) $Fe(s) + 2Ag^{+}(aq) \rightleftharpoons Fe^{2+}(aq) + 2Ag(s)$

3) $Zn(s) + Fe^{2+}(aq) \rightleftharpoons Fe(s) + Zn^{2+}(aq)$

Sol. 1) $E^{\circ}_{cell} = E^{\circ}_{O.P}$ of $Zn + E^{\circ}_{R.P}$ of Ag^{+}

$= (+0.76) + (0.80) = 1.56\,V$

2) $E^{\circ}_{cell} = E^{\circ}_{O.P}$ of $Fe + E^{\circ}_{R.P}\ Ag^{+}$

$= (0.44) + (0.80) = 1.24\,V$

3) $E^{\circ}_{cell} = E^{\circ}_{O.P}$ of $Zn + E^{\circ}_{R.P}$ of Fe^{+2}

$= 0.76 + (-0.44) = 0.32\,V$

Q3 **The standard reduction potentials of $Zn^{2+} \mid Zn$, $Cu^{2+} \mid Cu$ and $Ag^{+} \mid Ag$ are respectively -0.76, 0.34 and 0.8 V. The following cells were constructed.**

I. $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$ **II. $Zn \mid Zn^{2+} \parallel Ag^{+} \mid Ag$**

III. $Cu \mid Cu^{2+} \parallel Ag^{+} \mid Ag$

What is the correct of E°_{cell} of these cells ?

(A) II > III > I **(B) II > I > III** **(C) I > II > III** **(D) III > I > II**

Sol. **(B)**

I. $E^{\circ} = (+0.76) + (0.34) = 1.10\,V$

II. $E^{\circ} = (+0.76) + (0.8) = 1.56\,V$

III. $E^{\circ} = (-0.34) + (0.8) = 0.46\,V$

Q4 **Consider the following E° values $E^{\circ}_{Fe^{3+}/Fe^{2+}} = +0.77\,V$, $E^{\circ}_{Sn^{2+}/Sn} = -0.14\,V$**

Under standard conditions the potential for the reaction

$Sn_{(s)} + 2Fe^{3+}(aq) \rightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq)$ is

(A) 0.63 V **(B) 1.40** **(C) 0.91 V** **(D) 1.68 V**

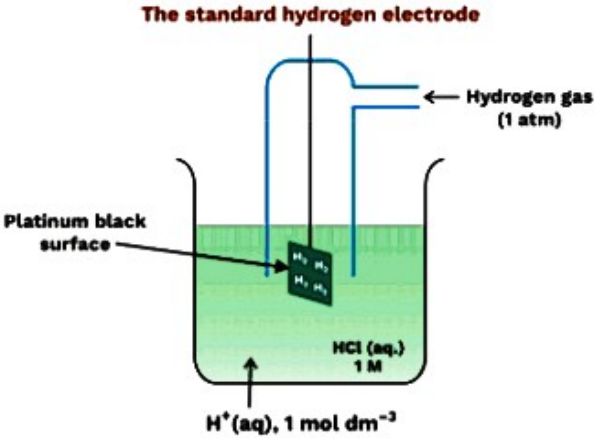
Sol. **(C)**

$$E^{\circ}_{cell} = E^{\circ}_{O.P} + E^{\circ}_{R.P}$$
$$= (+0.14) + (0.77) = 0.91\,V$$

$$E^{\circ}_{cell} \neq (E_{O.P.}\ \text{of Sn}) + 2\{E_{R.P}\ \text{of}\ Fe^{+3}\}$$

Electrode potential is an intensive property. Hence does not depends on moles.

Standard Hydrogen Electrode (S.H.E)



It is half cell of hydrogen in standard state.

H_2 gas is adsorbed on the surface of Pt-electrode (inert) hence an electrode of hydrogen is prepared.

Now, the half cell is prepared by placing this electrode in aqueous solution containing H^+ ions.

An inverted glass tube filled with hydrogen gas at standard conditions : Pressure of hydrogen = 1 atm

Pt plate (electrode) an inert electrode used to accumulate hydrogen gas on surface.

Concentration of $[H^+]$ = 1 M {standard condition}

Note:

Platinum Black, a type of platinum which contains more vacant sites, used to adsorb H_2.

$H_2(g)$ oxidation state of H = zero
$H^+(HCl)$ oxidation state of H = 1

It is a reversible half cell means can act as anode or cathode at a time.

Hence, hydrogen can act as anode as well as cathode.

a) Hydrogen as anode

Half-cell reaction (HCR) :

$$H_2(g) \longrightarrow 2H^+(aq) + 2e^-$$

Loss of electron or oxidation

Half-cell representation : $H_2(g)\,|\,Pt(s)\,|\,H^+(aq)$

b) Hydrogen as cathode

Half-cell reaction (HCR) :

$2e^- + 2H^+_{(aq)} \longrightarrow H_2(g)$

Gain of electron or reduction

Half-cell representation : $H^+(aq)\,|\,H_{2(g)}\,|Pt(s)$

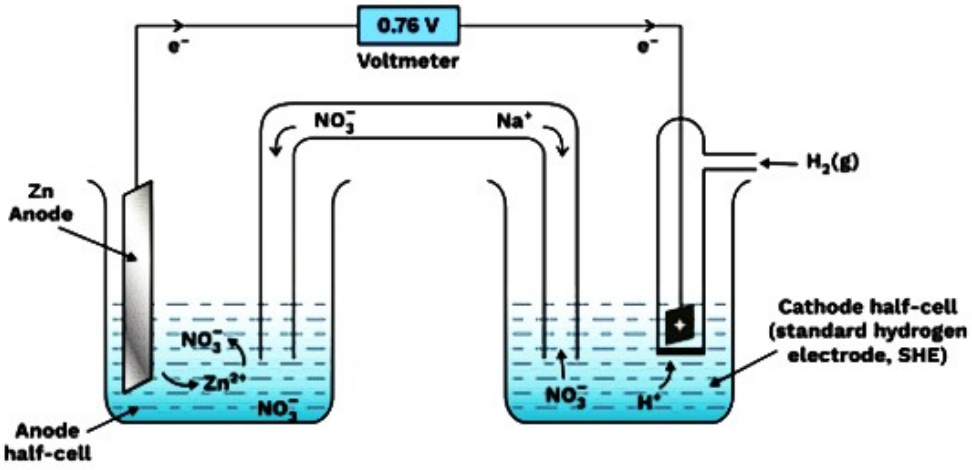
Points to remember

By IUPAC convention

$$E^\circ_{H_2/H^+} = E^\circ_{H^+/H_2} = 0.0 \text{ volt}$$

To determine E° of a substance, its half cell is connected with S.H.E. in standard conditions.

CASE-1 : If Zn half-cell is connected with S.H.E.



Since, we don't know Zn is cathode / anode.

To check, we remove Zn electrode from time to time.

Since size of zinc electrode decrease with time, it indicates that Zinc is being chipped off from its surface according to following reaction,

$$Zn(s) \longrightarrow Zn^{+2}(aq) + 2e^-$$

This concludes that Zinc gets oxidised as acts as anode

Consequently, Hydrogen gets reduced and same can be confirmed by evolution of a gas (H2 gas) from Hydrogen electrode side, according to flowing reaction,

$2H^+_{(ac)} + 2e^- \longrightarrow H_{2(g)}$

Hydrogen is reduced acts as cathode.

Internal circuit

$$\underset{\text{excess in anode}}{Zn^{+2}} + \underset{\text{(salt bridge)}}{2Cl^-} \longrightarrow ZnCl_2$$

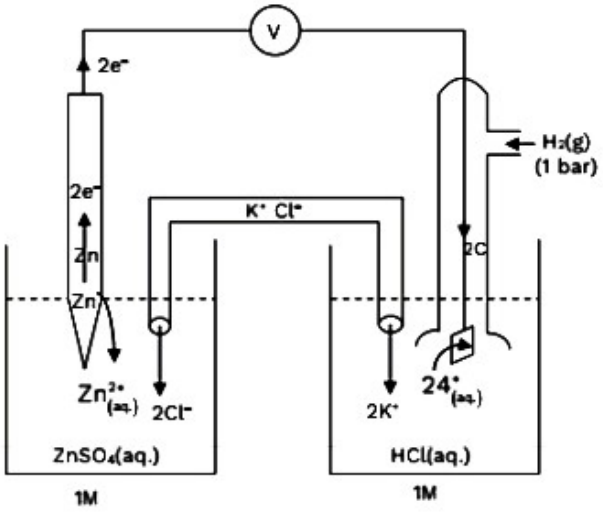
$$\underset{\text{(at cathode)}}{2Cl^-} + 2K^+ \longrightarrow 2KCl$$

External circuit $E^\circ_{cell} = 0.76\,V$ (experiment)

Anode $Zn(s) \rightleftharpoons Zn^{2+}_{(ac.)} + 2e^-$

Cathode $2H^+_{(ac.)} + 2e^- \rightleftharpoons H_2(s)$

NCR $\quad Zn(s) + 2H^+_{(aq.)} \rightleftharpoons Zn^{2+}_{(ac.)} + H_2(s)$



Cell representation :

$$Zn(s)\,|\,Zn^{2+}_{(aq.)}\,||\,H^+_{(aq.)}\,|H_{2(s)}\,|\,Pt\,(s)$$

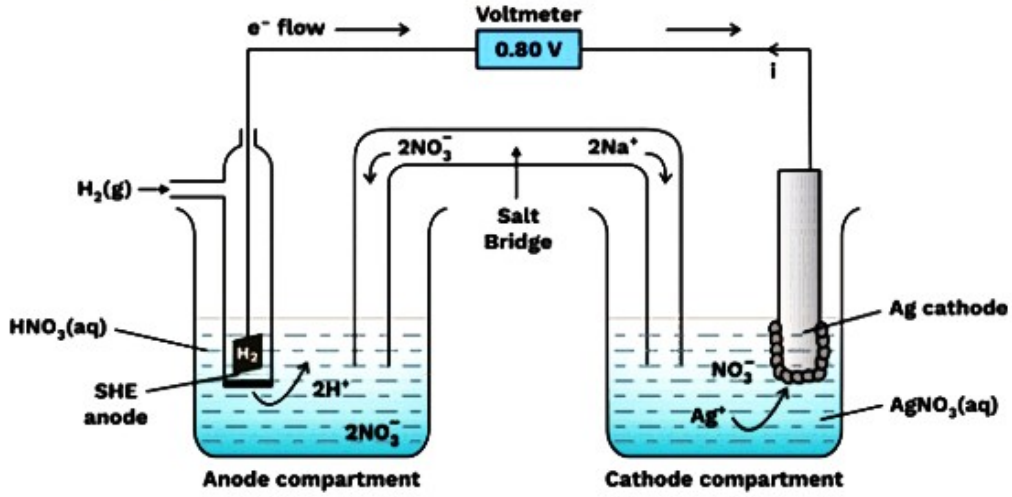
Now,

$$E^\circ_{cell} = E^\circ_{OP} + E^\circ_{RP}$$

$$0.76\,V = \left(E^\circ_{Zn/Zn^{2+}}\right) + \left(E^\circ_{H^+/H_2}\right)$$

$$\Rightarrow E_{Zn/Zn^{2+}} = 0.76\,V$$

CASE-2 : If Ag half-cell is connected with S.H.E



In this case when silver electrode is removed from time to time, an increase in mass of electrode is observed, this confirms deposition of Ag on the electrode surface according to the equation,

Hence, $Ag^{+}_{(aq)} + e^{-} \longrightarrow \underset{\text{electrode}}{Ag(s)}$

From above observation we can conclude that Silver electrode behaves as cathode and undergoes reduction.

At anode $H_2(g) \rightleftharpoons 2H^{+}_{(aq.)} + 2e^{-}$

At cathode $Ag^{+}_{(aq.)} + e^{-} \rightleftharpoons Ag(s)$

NCR $H_{2(g)} + 2Ag^{+}_{(aq.)} \rightleftharpoons 2H^{+}_{(aq.)} + 2Ag(s)$

Cell representation :

$$Pt(s)\,|\,H_{2(g)}\,|H^{+}_{(aq.)}\,||\,Ag^{+}_{(aq.)}\,|Ag(s)$$

Also, $0.8\,V = \left(E^{\circ}_{H_2/H^{+}}\right) + \left(E^{\circ}_{Ag^{+}/Ag}\right)$

$\Rightarrow E^{\circ}_{Ag^{+}/Ag} = 0.8\,V$

Example : $E^{\circ}_{Zn^{2+}/Zn} = -0.76\,V$

$E^{\circ}_{Fe^{2+}/Fe} = -0.44\,V$

$E^{\circ}_{Cu^{2+}/Cu} = 0.34\,V$

$E^{\circ}_{Ag^{+}/Ag} = 0.8\,V$ etc.

Points to remember

By IUPAC convention, E° of substances are generally expressed as SRP values.

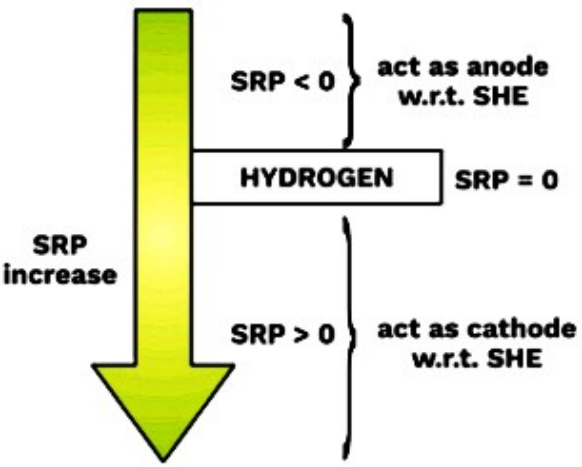
Electrochemical Series

So, In a galvanic cell,

Cathode → Greater SRP substance

Anode → Lesser SRP substance

To get oxidised or reduced is a relative tendency and reference of this relative tendency is compared with the help of electro chemical series. For instance, in Ag-Cu cell reduction potentials are as follow



$\because$ $E^{\circ}_{Ag^{+}/Ag} = 0.8\,V$

$4E^{\circ}_{Cu^{2+}/Cu} = 0.34\,V$

Higher reduction potential indicates Higher tendency to get reduced and vice versa.

Since, R.P of $Ag^{+} > Cu^{+2}$ Ag will get reduced and Cu will get oxidized and

hence,
Ag : cathode
Cu : anode
Representation of cell

$\Rightarrow$ $Cu(s)\,|\,Cu^{2+}_{(ac.)}\,\|\,Ag^{+}_{(aq.)}\,|\,Ag(s)$

Anode Cathode

More SRP value $\Rightarrow$	More tendency to be reduced $\Rightarrow$	More tendency to oxidise other $\Rightarrow$	Better oxidising agent

Increasing strength of reducing agent

Electrode	Reaction	SRP (at 298 K)
*Li	$Li^{+} + e^{-} \rightarrow Li(s)$	– 3.05 V
K	$K^{+} + e^{-} \rightarrow K(s)$	– 2.93 V
Ba		
Ca	$Ca^{+2} + 2e^{-} \rightarrow Ca(s)$	– 2.87 V
Na	$Na^{+} + e^{-} \rightarrow Na(s)$	– 2.71 V
Mg	$Mg^{+2} + 2e^{-} \rightarrow Mg(s)$	– 2.37 V
Al		
* Electrolytes (H_2O)	$H_2O(l) + e^{-} \rightarrow ½\,H_2 + OH^{-}$	– 0.828 V
*Zn	$Zn^{+2} + 2e^{-} \rightarrow Zn(s)$	– 0.76 V
Cr	$Cr^{+3} + 3e^{-} \rightarrow Cr(s)$	– 0.74 V
*Fe	$Fe^{2+} + 2e^{-} \rightarrow Fe$	– 0.44 V
Cd	$Cd^{+2} + 2e^{-} \rightarrow Cd(s)$	– 0.40 V
Co		
Ni	$Ni^{+2} + 2e^{-} \rightarrow Ni(s)$	– 0.24 V
Sn	$Sn^{+2} + 2e^{-} \rightarrow Sn(s)$	– 0.14 V
Pb	$Pb^{+2} + 2e^{-} \rightarrow Pb(s)$	– 0.13 V
*H_2	$2H^{+} + 2e^{-} \rightarrow H_2(g)$	0.00 V
Cu	$Cu^{2+} + 2e^{-} \rightarrow Cu(s)$	0.34 V
Fe	$Fe^{3+} + e^{-} \rightarrow Fe^{2+}$	0.77 V
Hg	$Hg_2^{2+} + 2e^{-} \rightarrow Hg(l)$	0.79 V
Ag	$Ag^{+} + e^{-} \rightarrow Ag$	0.80 V
Hg	$Hg^{2+} \rightarrow Hg(l)$	0.85 V
Br_2	$Br_2 + 2e^{-} \rightarrow 2Br^{-}$	1.06 V

Electrolytes $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O(l)$	1.23 V
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{+3} + 7H_2O$	1.33 V
$Cl_2 + 2e^- \rightarrow 2\,Cl^-$	1.36 V
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	1.51 V
$F_2 + 2e^- \rightarrow 2F^-$	2.87 V

Important Point

Since, electrode potential is a path intensive property, in a electrochemical reaction, applying Hess Law for E or E° is technically wrong. Hess Law is always applied of G or G° and by using relation between Gibb's free energy (G) and electrode potential (E) the final value of electrode or cell potential is calculated.

Q5 **Given $E^\circ_{Cr^{3+}/Cr} = -0.74\,V$; $E^\circ_{MnO_4^-/Mn^{2+}} = 1.51V$**

$E^\circ_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\,V$; $E^\circ_{Cl/Cl^-} = 1.36\,V$

Based on the data given above, strongest oxidising agent will be

(A) Cr^{3+} **(B) Mn^{2+}** **(C) MnO_4^-** **(D) Cl^-**

Sol. **(C)**

Strongest oxidising agent i.e., highest tendency to get reduced.

All data is S.R.P values

Higher S.R.P is strongest oxidising agent

Q6 **Given $E^\circ_{Cl_2/Cl^-} = 1.36\,V$, $E^\circ_{Cr^{3+}/Cr} = -0.74\,V$**

$E^\circ_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\,V$, $E^\circ_{MnO_4^-/Mn^{2+}} = 1.51V$

Among the following, the strongest reducing agent is

(A) Mn^{2+} **(B) Cr^{3+}** **(C) Cl^-** **(D) Cr**

Sol. **(D)**

Strongest reducing agent means maximum tendency is get oxidised.

All data S.R.P i.e., Max S.O.P is min S.R.P ($E_{Cr^{+3}/Cr}$)

Ans. (D) not (B) because Cr gets oxidised.

Q7 The standard reduction potentials for Zn^{2+}/Zn, Ni^{2+}/Ni, and Fe^{2+}/Fe are -0.76, -0.23 and -0.44 V respectively. The reaction $X + Y^{2+} \rightarrow X^{2+} + Y$ will be spontaneous when

(A) X = Fe, Y = Zn (B) X = Zn, Y = Ni (C) X = Ni, Y = Fe (D) X = Ni, Y = Zn

Sol. (B)

Q8 For the reduction of NO_3^- ion in an aqueous solution, E° is +0.96 V. Values of E° for some metal ions are given below

$V^{2+}(aq) + 2e^- \rightarrow V$ $\quad E^\circ = -1.19\,V$

$Fe^{3+}(aq) + 3e^- \rightarrow Fe$ $\quad E^\circ = -0.04\,V$

$Au^{3+}(aq) + 3e^- \rightarrow Au$ $\quad E^\circ = +1.40\,V$

$Hg^{2+}(aq) + 2e^- \rightarrow Hg$ $\quad E^\circ = +0.86\,V$

The pair(s) of metal that is (are) oxidised by NO_3^- in aqueous solution is (are)

(A) V and Hg (B) Hg and Fe (C) Fe and Au (D) Fe and V

Sol. (ABD)

Q9 If for the half cell reactions :

$Cu^{2+} + e^- \longrightarrow Cu^+$; $E^\circ = 0.15\,V$ $\quad Cu^{2+} + 2e^- \longrightarrow Cu$; $E^\circ = 0.34\,V$

Calculate E° of the half cell reaction $Cu^+ + e^- \longrightarrow Cu$

(A) 0.53 V (B) 0.49 V (C) 0.34 V (D) 0.19 V

Sol. (A)

Q10 Given below are the half-cell reaction :

$Mn^{2+} + 2e^- \rightarrow Mn$; $E^\circ = -1.18\,V$ $\quad 2(Mn^{3+} + e^- \rightarrow Mn^{2+})$; $E^\circ = +1.51\,V$

The E° for $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$ will be

(A) –2.69 V ; the reaction will occur
(B) –0.33 V ; the reaction will not occur
(C) –0.33 V ; the reaction will occur
(D) –2.69 V ; the reaction will not occur

Sol. (D)

NERNST Equation

The dependence upon the concentration can be derived from thermodynamics.

From thermodynamics

$\Delta G = \Delta G^\circ + RT \ln Q$

$-nFE = -nFE^\circ + 2.303\ R\ T \log Q$

$E = E^\circ - \frac{2.303RT}{nF} \log Q$

Take $T = 298$ K , $R = 8.314$ J/mol K ,

$F = 96500$ C

Now we get, $E = E^\circ - \frac{0.059}{n} \log Q$

Where n = number of transferred electron,

Q = reaction quotient

Do you Know?

Cell potentials depend on temperature and on the composition of the reaction mixtures.
It depends upon the concentration of the solute and the partial pressure of the gas, if any.

Points to remember

Nernst equation can be used to calculate cell potentials for nonstandard conditions also.
Nernst equations can be applied to half-cell reactions also

Applications of Nerst equation

Nernst Equation for Electrode Potential

$M^{n+}(aq) + ne^- \rightleftharpoons M(s)$

$$E_{Redn} = E^0_{red} - \frac{RT}{nF} \ln \left[\frac{M(s)}{M^{n+}}\right]$$

$$E_{Redn} = E^0_{red} - \frac{2.303RT}{nF} \log \left[\frac{M(s)}{M^{n+}}\right]$$

at 298K,

$$E_{Redn} = E^0_{Red^n} - \frac{0.059}{n} \log \left[\frac{1}{M^{n+}}\right]$$

Hydrogen Electrode

$H_2(g) \rightleftharpoons 2H^+(aq) + 2e^-$

$$E = E^0 - \frac{0.0591}{2} \log \left[\frac{(H^+)^2}{P_{H_2}}\right]$$

Metal–metal soluble salt electrode

$Zn^{2+} + 2e^- \rightarrow Zn(s)$

$$E_{Redn} = E^0_{Red^n} - \frac{2.303RT}{nF} \log \left(\frac{1}{Zn^{+2}}\right) \text{ at 298K}$$

$$E_{Redn} = E^0_{Red^n} - \frac{0.059}{2} \log \left(\frac{1}{Zn^{+2}}\right)$$

Gas – electrode
Hydrogen electrode

$$2H^+ + 2e^- \rightarrow H_2(g)$$

$$E_{Redn} = E^\circ_{red} - \frac{0.059}{2} \log\left(\frac{P_{H_2}}{[H^+]^2}\right)$$

Redox electrode

$$4H_2O + Mn^{2+} \rightarrow MnO_4^- + 8H^+ + 5e^-$$

$$E_{Ox} = E^\circ_{Ox} - \frac{0.059}{5} \log \frac{[MnO_4^-][H^+]^8}{[Mn^{+2}]}$$

Nernst Equation for cell Potential :

$aA + bB \rightleftharpoons cC + dD$

$$E_{cell} = E^0_{cell} - \frac{RT}{nF} \ln Q$$

n – no. of electrons which gets cancelled out while making cell reaction.

Equilibrium in electrochemical cell

$\Delta G^0 = - nF E^\circ_{cell}$

$\Delta G = - nF E_{cell}$

From thermo dynamics

$\Delta G = \Delta G^0 + RT\ln Q$

so, $\Delta G^0 = - RT\ln K_{eq}$

at equilibrium $- nF E^\circ_{cell} = -2.303 RT \log (K_{eq})$

$$\log K_{eq} = \frac{nF}{2.303RT} E^\circ_{cell}$$

at 298 K and R = 8.314 J/mol K

$$\log K_{eq} = \frac{n}{0.059} E^\circ_{cell}$$

Points to remember

at chemical equilibrium $\Delta G = 0$
$E_{cell} = 0 \rightarrow$ cell will be of no use

Q11 **The standard e.m.f. of a cell involving one electron change is found to be 0.59IV at 25°C. The equilibrium constant of the reaction is F=96,500C mol^{-1}; R = 8.31.4 $JK^{-1}mol^{-1}$.** **(AIEEE 2004)**

(A) 1.0×10^{30} **(B) 1.0×10^5** **(C) 1.0×10^{10}** **(D) 1.0×10^1**

Sol. **(C)**

$E^\circ_{cell} = 0.591V$ $\Delta G^\circ = -nFE^\circ_{cell} = -RT\ln k_{eq}$

T = 298 K at 298 Kelvin

$n = 1$ (no. of electrons) $\Rightarrow E^{\circ}_{cell} = \frac{0.059}{n}\log K_{eq}$

$0.59 = \frac{0.05}{1}\log K_{eq}$

$\text{Log } K_{eq} = 10$

$\Rightarrow \quad K_{eq} = 10^{10}$

Q12 **Given $E^{\circ}_{Cr^{3+}/Cr} = -0.72V$; $E^{\circ}_{Fe^{2+}/Fe} = -0.42V$**

The potential for the cell Cr / Cr^{3+} (0.1M) || Fe^{2+} (0.01M)| Fe is:

(A) -0.339V **(B) −0.26V** **(C) 0.26V** **(D) 0.339**

Sol. **(C)**

Anode reaction $Cr(s) \rightleftharpoons Cr^{3+}_{(eq)} + 3e^{-}(\text{oxidation})$

Cathode reaction $2e^{-} + Fe^{2+}_{(eq)} \rightleftharpoons Fe(s)(\text{Pr oduction})$

$2Cr(s) + 3Fe^{2+}_{(eq)} \rightleftharpoons 2Cr^{3+}_{(eq)} + 3E(s); n = 6$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.059}{n}\log\frac{\left[Cr^{3+}_{(eq)}\right]^2}{1.\left[Fe^{2+}_{(eq)}\right]^3}$$

$$= (0.3) - \frac{0.06}{6}\log\frac{(0.1)^2}{(0.01)^3}$$

$$= 0.3 - 0.01\log\frac{10^{-2}}{10^{-6}}$$

$= 0.3 - 0.04 = 0.26V$

Q13 **The reduction potential of hydrogen half-cell will be negative if**

(A) $P(H_2)$=1 atm and $[H^+]$ = 2.0M **(B) $P(H_2)$=1 atm and $[H^+]$ = 1.0M**

(C) $P(H_2)$=2 atm and $[H^+]$ = 1.0M **(D) $P(H_2)$=2 atm and $[H^+]$ = 2.0M**

Sol. **(C)**

For hydrogen half cell

Reduction $\Rightarrow 2e^{-} + 2H^{+}_{(eq)} \rightleftharpoons H_2(j); (n = 2)$

$$E_{RP} = E^{\circ}_{R^{\circ}} - \frac{0.051}{n}\log\frac{[P_{H_2}]}{[H^+]^2}$$

For hydrogen $E^{\circ}_{cell} = -\frac{0.059}{2}\log\frac{P_{H_2}}{(H^+)^2}$

For E_{RP} to be negative (all is ques)

$\log\frac{P_{H_2}}{(H^+)^2} > 0$ (positive) ie $\frac{P_{H_2}}{(H^+)^2} > 1$

All option possible for option C.

Work done by a cell

(i) Let 'n' faraday charge be taken out of a cell of EMF 'E' ; then work done by the cell will be calculated as :
Work = Charge × Potential = nFE

(ii) Work done by cell = Decrease in free energy
so $-\Delta G = nFE$
or $W_{max} = +nFE^{\circ}$ where E° is standard EMF of the cell

Q14 **For the electrochemical cell, Mg(s) | Mg^{2+} (aq.1M) || Cu^{2+} (aq.1M) | Cu(s)**

The standard emf of the cell is 2.70V at 300K. When the concentration of Mg^{2+} is changed to xM, the cell potential changes to 2.67V at 300K. The value of x is _____.

(Given, $\frac{F}{R} = 11500KV^{-1}$, where F is the Faraday constant and R is the gas constant. ln(1)=2.30).

Sol. **(i)**

If Cu^{+2} is kept at 1M

Anode : $Mg(s) \rightleftharpoons Mg^{2+}_{(eq)} + 2e^-$

Cathode : $Cu^{2+}_{(eq)} + 2e^- \rightleftharpoons Cu(s)$

NCR $Mg(s) + Cu^{2+}_{(eq)} \rightleftharpoons Mg^{2+}_{(eq)} + Cu(s)$ [n = 2]

$$E = E^{\circ} - \frac{RT}{nF}\ln\frac{\left[Mg^{2+}\right]^1}{\left[Cu^{2+}\right]^1}$$

$$2.67 = 2.7 - \frac{300}{11500 \times 2}\log\frac{x}{1}$$

$\left[Cu^{+2}\right]$ = 1M{given}

$$\frac{0.03\times11500\times2}{300}=\ln x$$

$2.3 = \ln x \Rightarrow x = 10$

Sol. **(ii)**

Concentration of Mg^{+2} changes to x due to cell reaction. Here concentration of Cu^{+2} decreases.

$$Mg(s) + Cu^{2+}_{(aq)} \rightleftharpoons Mg^{2+}_{(aq)} + Cu(s)$$

t = 0	1 M	1M
t = t	1–y	(1+y) = x

Now $2.67 = 2.7 - \frac{RT}{nF}\ln\frac{x}{2-x}$

$$\Rightarrow \frac{x}{2-x} = 10$$

$x = 20 - 10x$

$$x = \frac{20}{11}$$

Q15 **For the following cell $Zn(s) \mid ZnSO_4(aq.) \parallel CuSO_4 \mid Cu(s)$**

When the concentration of Zn^{2+} is 10 times the concentration of Cu^{2+}, the expression for $\Delta G(inJmol^{-1})$ is

[F is Faraday constant; R is gas constant; T is temperature E°(cell) = 1.1V]

(A) 2.303 RT – 2.2 F **(B) 2.303 RT + 1.1 F**

(C) 1.1 F **(D) –2.2 F**

Sol. **(A)**

$\Delta G° = nFE°$ (as E° given in question)

$\Delta G = \Delta G + RT\ln Q$

$=(-2F \times 1.1) + RT \ln 10$

$= 0.303\ RT = 0.303\ RT - 2.2\ F$

Q16 **For the following electrochemical cell at 298K,**

$Pt(s) \mid H_2(g.1bar) \mid H^+(aq.1M) \mid\mid M^{4+}(aq), M^{2+}(aq) \mid Pt(s)$; $E_{cell} = 0.092V$ when $\left[\frac{M^{2+}(aq)}{M^{4+}(aq)}\right] = 10^x$. Given $E^0_{M^{4+}/M^{2+}} = 0.151V : 2.303\frac{RT}{F} = 0.059V$

The value of x is

(A) –2 **(B) –1** **(C) 1** **(D) 2**

Sol. **(D)**

$$H_2(O) \rightleftharpoons 2H^+_{(eq)} + 2e^-$$

$$M^{4+}_{(eq)} + 2e^- \rightleftharpoons M^{2+}_{(eq)}$$

$$H_2O) + M^{4+}_{(eq)} \rightleftharpoons 2H^+_{(eq)} + M^{2+}_{(eq)}$$

$$E = E^\circ - \frac{0.059}{n}\log\frac{[H^+]^2\left[M^{2+}\right]^1}{\left(P_{H_2}\right)^1\left[M^{4+}\right]^1}$$

$$0.092 = (0 + 0.151) - \frac{0.059}{2}\log\frac{(1)^2.10^x}{1}$$

On solving x = 2

In Formula : $E = E^\circ - \frac{RT}{nF}\ln Q$

$$= E^\circ - \frac{0.059}{n}\log Q$$

Q = equilibrium constant of net cell reaction

Q17 **Consider the following cell reaction :**

$$\overset{\bullet}{2F}e(s) + \overset{\bullet}{O}_2(g) + 4H^+(aq) \longrightarrow 2\overset{2+}{Fe}^{2+}(aq) + 2H_2\overset{-2}{O}(l) \quad E^\circ = 1.67\,V$$

At $[Fe^{2+}] = 10^{-3}$ M, $P(O_2) = 0.1$ atm and pH = 3, the cell potential at 25°C is

(A) 1.47 V **(B) 1.77 V** **(C) 1.87 V** **(D) 1.57 V**

Sol. **(D)**

Cell reaction: $2Fe(s) \rightarrow 2Fe^{+2} + 4e^-$ (Oxidation Half Reaction)

$4e^-O_2(g) \rightarrow 2H_2O^{-2}$ (Reduction Half Reaction)

$$4H^{+} + 2Fe(s) + O_2(g) \rightarrow 2Fe^{+2}(g) + 2H_2O$$

Cell Reaction n (No. of electrons transferred) = 4

$$E = E^{\circ} - \frac{0.059}{n}\log\frac{[Fe^{2+}]^2 \cdot 1}{1 \cdot (P_{O_2})^1[H^+]^4}$$

$$E = (1.67) - \frac{0.06}{4}\log\frac{(10^{-3})^2}{(0.1)^1(10^{-3})^4} = 1.67 - 0.1 = 1.57\,V$$

Q18 **For the cell, Pt(s) | H_2(g, 1 atm) | HA(aq, 0.1 M) || Ag^+ (aq, 10^{-8} M) | Ag(s); if E_{cell} at 25°C is 0.50 V then the value of ionisation constant of the acid HA is, [Given : $E^{\circ}(Ag^+ | Ag) = 0.8$ V]**

(A) 1.0×10^{-5} (B) 8.0×10^{-3} (C) 2.5×10^{-5} (D) 3.2×10^{-6}

Sol. **(A)**

Anode : $\frac{1}{2}H_2(s) \rightleftharpoons H^+_{(aq.)} + e^-$

Cathode : $Ag^+_{(aq.)} + e^- \rightleftharpoons Ag(s)$

Nett Cell Reaction: $\frac{1}{2}H_2(s) + Ag^+_{(aq.)} \rightleftharpoons H^+_{(aq.)} + Ag(s)$

$$E = E^{\circ} - \frac{0.059}{n}\log\frac{[H^+]}{(P_{Cl_2})^{1/2}[Ag^+]^1}$$

$$0.5 = (0 + 0.8) - \frac{0.06}{1}\log\frac{[H^+]}{(1)^{1/2}(10^{-8})^1}$$

$$-0.3 = -0.06\log\frac{[H^+]}{(10^{-8})}$$

$$[H^+] = 10^{-3}M$$

To calculate the value of ionisation :

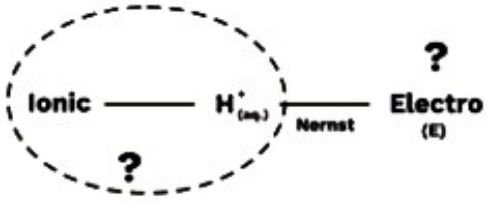
Applying the concept of ionic equilibrium (ostwald's law)

K_a of acid if its 0.1 M solution has

$[H^+] = 10^{-3}$ M

Points to remember

Ionic and electrochemistry , both can have inter-related question . Like, in above question pH is given, convert it into [H+] ions and then applying Nernst equation for applying electrochemistry concept. And vice versa is also possible.



	$HA_{(aq.)}$	$\rightleftharpoons H^+_{(aq.)}$	+	$A^-_{(aq.)}$
t = 0	0.1M		—	—
t_{eq}	0.1 – x		$x = 10^{-3}_{M}$	x
	≈ (0.1)			

$$K_{eq} \Rightarrow K_a = \frac{(10^{-3})(10^{-3})^1}{(0.1)^1}$$

$$K_a = 10^{-5}$$

Q19 **Consider an electrochemical cell : $A(s) \mid A^{n+}(aq, 2M) \mid\mid B^{2n+}(aq, 1M) \mid B(s)$. The values of ΔH^θ for the cell reaction is twice that of ΔG^θ at 300 K. If the emf of the cell is zero, the ΔS^θ (in $JK^{-1}\ mol^{-1}$) of the cell reaction per mole of B formed at 300 K is _____.**
(Given : ln(2) = 0.7, R(universal gas constant) = 8.3 $JK^{-1}\ mol^{-1}$, H, S and G are enthalpy, entropy and Gibbs energy, respectively)

Sol. Cell reactions:

$$\left.\begin{array}{l} A_{(s)} \longrightarrow A^{n+}_{(aq.)} + ne^- \end{array}\right\} \times 2$$

$$\left.\begin{array}{l} B^{2n+}_{(aq.)} + 2ne^- \longrightarrow 8_{(s)} \end{array}\right\} \times 1$$

$$2A_{(s)} + B^{2n+}_{(aq.)} \longrightarrow 2A^{n+}_{(aq.)} + B_{(s)}$$

Number of electrons $e^- = (2n)$

$$\Delta G^\circ = \Delta H^\circ - T \cdot \Delta S^\circ$$

$$\Delta S^\circ = \frac{\Delta G^\circ}{T} = \left(\frac{-nFE^\circ}{T}\right)$$

Given: If the emf of the cell is zero

$$E = E^\circ - \frac{RT}{nF} \ln \frac{[A^{n+}]^2}{[B^{2n+}]^1}$$

$$(nFE^\circ) = RT \ln (2)^2 = -\Delta G^\circ$$

$$\Delta S^\circ = \frac{-RT(2 \ln 2)}{T}$$

$$= -2 \times 8.3 \times 0.7 \text{ J / K-mol}$$

Points to remember

Factors affecting E_{cell} :
i) Substances of anode of cathode
ii) Temperature
iii) Concentration of aq. Components
iv) Partial properties of gaseous components

Efficiency of a cell :

$$= \frac{\text{Electrical work doen by cell of constant Temperature \& Pressure}}{\text{Heat released by reaction at constant Temperature \& Pressure}}$$

i.e., $\boxed{\text{Efficiency} = \left|\frac{\Delta G}{\Delta H}\right|}$ if both in same units

Concentration cell :

These are galvanic cells in which both anode & cathode are of same substance.

i.e., for such cells :

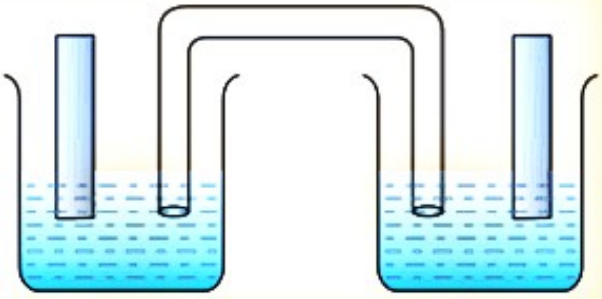
Hence $\boxed{E^{\circ}_{cell} = 0}$

The cell reaction takes place due to difference in concentrations of electrolyte in both compartments.

Q20 **Calculate the EMF of the following cell**

$Zn \mid Zn^{2+}(0.01M) \parallel Zn^{2+}(0.1M) \mid Zn$

at 298 K.

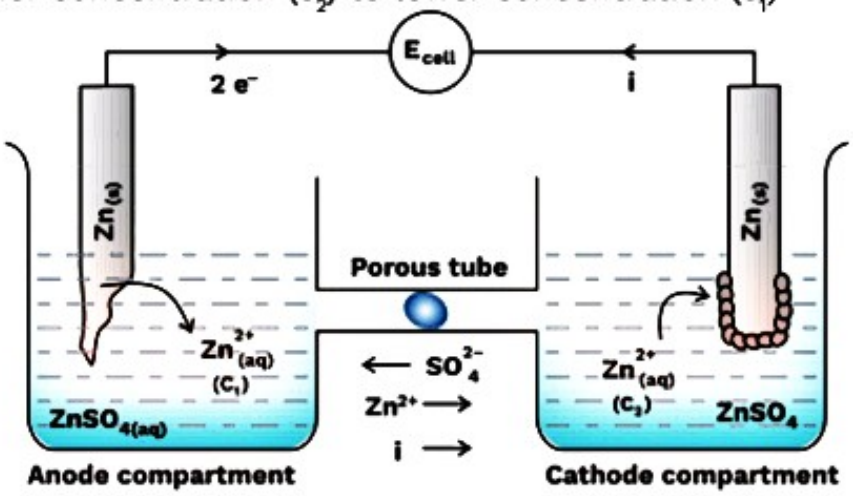


Sol. Anode: $Zn_{(s)} \rightleftharpoons Zn^{2+}_{(aq.)} + 2e^-$

Cathode: $Zn^{2+}_{(aq.)} + 2e^- \rightleftharpoons Zn(s)$

Nett Cell Reaction: $\underset{(c_2)}{Zn^{2+}_{(aq.)}} \rightleftharpoons \underset{(c_1)}{Zn^{2+}_{(aq.)}}$

From higher concentration (c_2) to lower concentration (c_1)



Working of porous tube: if charges are same in both compartment, no use.
But if concentration of ions varies, this will help in migration of ions.
In anode compartment (left): $[Zn^{+2}]$ is low so more formation of $[Zn^{+2}]$ takes place.
Hence, amount of electrode decreases.
In cathode compartment (right): $[Zn^{+2}]$ is high so more formation of [Zn] takes place.
Hence, amount of electrode increases.
Deposition of Zn takes place on electrode
In external circuit: electrons flow from anode to cathode.
In internal circuit :

At anode : less $[SO_4^{-2}]$

At cathode : more $[SO_4^{-2}]$

Hence, $[SO_4^{-2}]$ flows from cathode to anode.

Q21 $\mathbf{Ag(s)\left|AgCl(s)\right|Cl^-_{(aq.)}\left\|Br^-_{(aq.)}\right|AgBr(s)\left|Ag(s)\right.}$

$\mathbf{\Rightarrow Ag(s)\left|Ag^+_{(aq.)}\right\|Ag^+_{(aq.)}\left|Ag(s)\right.}$

$$\mathbf{E = 0 - \frac{0.059}{1}\log\frac{(K_1/C_1)^1}{(K_2/C_2)}}$$

Sol. Now $E = E^\circ_{cell} - \frac{RT}{nF}\ln\frac{(C_1)^1}{(C_2)^1}$

$$E = -\frac{RT}{nF}\ln\frac{C_1}{C_2}$$

For spontaneous cell:

$E_{cell} > 0 \Rightarrow \boxed{C_1 < C_2}$

Points to remember

Lower concentration electrode will behave as anode compartment and higher concentration cell as cathode compartment, in a concentration cell.

Q22 **For the cell $Pt\,|\,H_2(0.4\,atm)\,|\,H^+(pH=1)\,||\,H^+(pH=2)\,|\,H_2(0.1atm)\,|\,Pt$**

The measured potential at 25°C is

(A) −0.1 V **(B) −0.5** **(C) −0.041** **(D) None of these**

Sol. **(C)**

Anode : $\underset{(0.4\,atm)}{H_{2(s)}} \rightleftharpoons \underset{(0.1M)}{2H^+_{(aq.)}} + 2e^-$

Cathode : $\underset{(10^{-2}M)}{2H^+_{(aq.)}} + 2e^- \rightleftharpoons \underset{(0.1atm)}{H_{2(s)}}$

Nett Cell Reaction: $\underset{(0.4\,atm)}{H_{2(s)}} + \underset{(10^{-2}M)}{2H^{+}_{(ac.)}} \rightleftharpoons \underset{(0.1\,M)}{2H^{+}_{(ac.)}} + \underset{(0.1atm)}{H_{2(s)}}$

Applying Nernst Equation :

$$\text{Now } E_{cell} = 0 - \frac{0.059}{2}\log\frac{(0.1)^2(0.1)^1}{(0.4)^1(10^{-2})^2}$$

$$= -0.03\log\frac{10^{-3}}{4\times10^{-5}} = -0.03\times(2\log 5) = -0.042\,V$$

Non spontaneous cell.

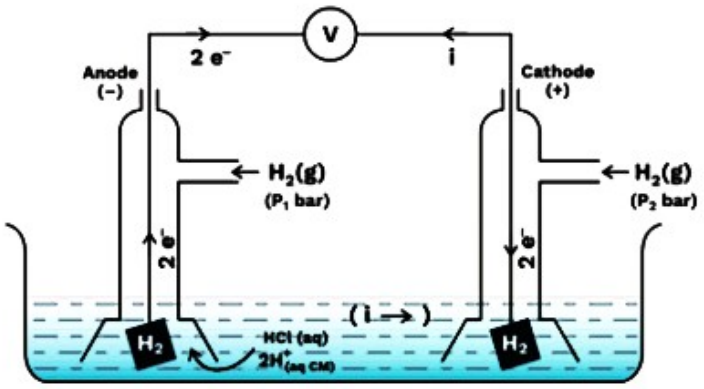
Q23 **Determine emf of following cell :**

$Pt(s)\,|\,H_2(g,1atm)\,|\,HCl(aq,0.1M)\,|\,H_2(g,0.1atm)\,|\,Pt(s)$

Sol. $Pt(s)\left|H_2(g,1\,atm)\right|HCl(aq.0.1\,M)\left|H_2(g,0.1\,atm)\right|Pt(s)$

$$\left.\begin{array}{l} H_2(g) \rightarrow H^{+}\{HCl\} \\ H^{+}\{HCl\} \rightarrow H_2(g) \end{array}\right\} \text{same container}$$

Note: Double bar is missing in above representation.
Cathode and anode in same container. No porous tube present.



Anode : $\underset{(1)}{H_{2(s)}} \rightleftharpoons \underset{(0.1)}{2H^{+}_{(ac.)}} + 2e^{-}$

Cathode : $\underset{(0.1M)}{2H^{1}_{(aq)}} + 2e^{-} \rightleftharpoons \underset{(0.1)}{H_2(s)}$

Nett Cell Reaction : $\underset{(1atm)}{H_{2(s)}} \rightleftharpoons \underset{(0.1atm)}{H_{2(s)}} \quad (e^{-} = 2)$

Applying Nernst equation:

$$E_{cell} = 0 - \frac{0.06}{2}\log\frac{(0.1)^1}{(1)^1} = 0.03\,V$$

In short:

Concentration cells

$$E^{\circ}_{cell} = 0$$

- Mixed questions of electrochemistry and ionic equilibrium are possible. To solve we will calculate the [H+] ions and apply Nernst equation or vice versa.

Electro ——— H^+ ——— **Ionic**

$$E = E^{\circ} - \frac{0.059}{5} \log \frac{[MnO^-_{4(aq.)}]^1 [H^+_{(aq.)}]^6}{[Mn^{2+}_{(aq.)}]^1}$$

Different types of half-cells

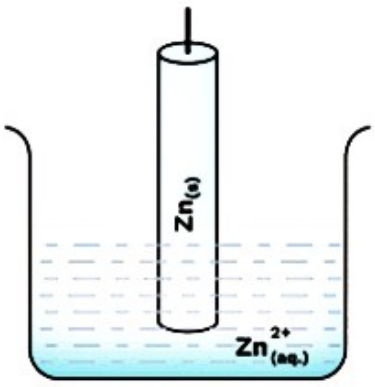
(1) 'Metal-ion' half-cell

An electrode of the metal is placed in an aqueous solution containing ions of same metal. For example, half cell of Zn, Cu, Fe, Ag etc. in aq. solution of their ions.

Half cell can act as anode as well as cathode, both. Representation of electrode will be different in both the cases.

IUPAC Representation as anode : $Metal_{(s)} \mid ion_{(aq.)}$

IUPAC Representation as cathode : $ion_{(aq.)} \mid Metal_{(s)}$



Example : Zn Half-cell

As anode :	**As cathode :**
Representation $Zn(s) \mid Zn^{2+}_{(aq.)}$	Representation $Zn^{2+}_{(aq.)} \mid Zn(s)$
Half Cell Reaction $\because$ oxidation	Half Cell Reaction $\because$ reduction
i.e., $Zn(s) \rightleftharpoons Zn^{2+}_{(aq.)} + 2e^-$	i.e., $2e^- + Zn^{2+}_{(aq)} \rightleftharpoons Zn(s)$

Note :

Hg is a liquid metal so for its half-cell inert electrode (Pt) is used.

As anode :

$Pt(s) \mid Hg(l) \mid Hg^{2+}_{2(aq.)}$

i.e., $2Hg(l) \rightleftharpoons Hg^{2+}_{2(aq.)} + 2e^-$

As cathode :

$Hg^{2+}_{2(aq.)} \mid Hg(l) \mid Pt(s)$

i.e., $Hg^{2+}_{2(aq.)} + 2e^- \rightleftharpoons 2Hg(l)$

(2) 'Gas-gas ion' half-cell

The gas is absorbed on the surface of inert electrode (Pt), & then the electrode is placed in aq. solution containing ions of same gas.

Example : Standard Hydrogen Electrode or Hydrogen Electrode

Representation as anode: $Pt(s) \mid Gas(s) \mid ion(aq.)$

Representation as cathode: $ion(aq.) \mid Gas(s) \mid Pt(s)$

Example : Chlorine half-cell (Cl_2 & Cl^-)

As anode :

$Pt(s) \mid Cl_{2(g)} \mid Cl^-_{(aq.)}$

Half Cell Reaction

$\because$ oxidation

i.e., $2Cl^-_{(aq.)} \rightleftharpoons Cl_{2(g)} + 2e^-$

As cathode :

$Cl^-_{(aq.)} \mid Cl_{2(g)} \mid Pt(s)$

Half Cell Reaction

$\because$ reduction

i.e., $Cl_{2(g)} + 2e^- \rightleftharpoons 2Cl^-_{(aq.)}$

Example : Oxygen half-cell (O_2 & OH^-)

As anode :

$Pt(s) \mid O_{2(g)} \mid OH^-_{(aq.)}$

Half Cell Reaction

$OH^-_{(aq.)} \rightleftharpoons O_{2(g)}$

$+H_2O$

$+3OH^- \quad +3H_2O$

$+4e^-$

$4OH^-_{(aq.)} \rightleftharpoons O_{2(g)} + 2H_2O + 4e^-$

As cathode :

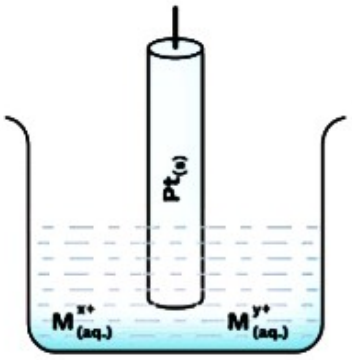
$OH^-_{(aq.)} \mid O_{2(g)} \mid Pt(s)$

Half Cell Reaction

$O_2(s) + 2H_2O_{(l)} + 4e^- \rightleftharpoons 4OH^-_{(aq.)}$

(3) 'Ion-Ion' half-cell

An inert electrode is placed in an aqueous solution of a substance containing two different types of ions of same substance.



Representation as anode:

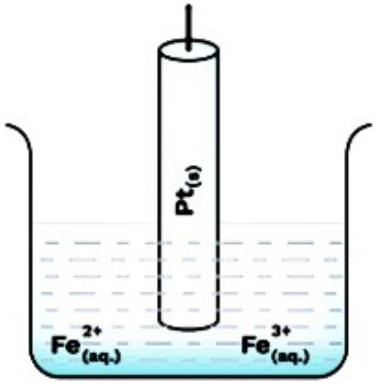
$$Pt(s) \mid M^{x+}_{(aq.)}, M^{y+}_{(aq.)} \quad (\text{for } x < y)$$

Representation as cathode:

$$M^{x+}_{(aq.)}, M^{y+}_{(aq.)} \mid Pt(s) \quad (\text{for } x > y)$$

Two different oxidation states of ions 'x' and 'y'.

Example: Cell of Fe^{2+} and Fe^{3+}



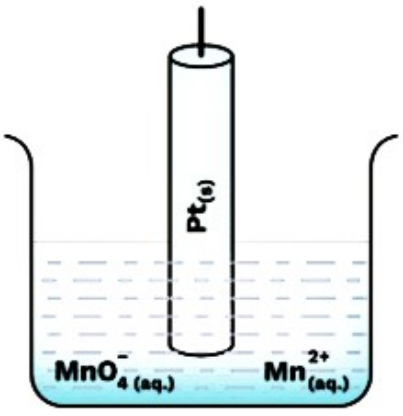
As anode :	**As cathode :**
$Pt(s) \mid Fe^{2+}_{(aq.)}, Fe^{3+}_{(aq.)}$	$Fe^{3+}_{(aq.)}, Fe^{2+}_{(aq.)} \mid Pt(s)$
Half Cell Reaction	Half Cell Reaction
∵ oxidation	∵ reduction
i.e., $Fe^{2+}_{(aq.)} \rightleftharpoons Fe^{3+}_{(g)} + e^-$	i.e., $Fe^{3+}_{(aq.)} + e^- \rightleftharpoons Fe^{2+}_{(aq)}$

Example: Cell of MnO_4^- and Mn^{2+}

As anode :

Half Cell Reaction

∵ oxidation



i.e., $Mn^{2+}_{(aq.)} \rightleftharpoons MnO^-_{4(aq.)}$

$+4H_2O(l) \qquad +8H^+_{(aq.)}$

$+5e^-$

$$Mn^{2+}_{(aq.)} + 4H_2O(l) \rightleftharpoons MnO^-_{4(aq.)} + 8H^+_{(aq.)} + 5e^-$$

Representation $Pt(s) \mid MnO^-_{4(aq.)}, Mn^{2+}_{(aq.)}, H^+_{(aq.)}$

Applying Nernst equation:

$$E = E^\circ - \frac{0.059}{5} \log \frac{[MnO^-_{4(aq.)}]^1 [H^+_{(aq.)}]^8}{[Mn^{2+}_{(aq.)}]^1}$$

In short

$\Delta G = -nfE_{cell}$ If $E > 0 \Rightarrow$ Spontaneous

$E < 0 \Rightarrow$ Spontaneous

$E = 0 \Rightarrow$ Equilibrium

E & $E^\circ \rightarrow$ Intensive properties

(4) Metal-insoluble salt' half cell Or 'Metal-sss-anion' half cell

To Maintain the potential of a half cell, it is necessary that concentrations of all ions involved in the cell reaction remains constant.

For instance, consider $Zn(s)\left|zn^{2+}_{(aq.)}\right.$ half cell

Half-cell Reaction:

$$Zn(s) \rightleftharpoons Zn^{2+}_{(aq)} + 2e^-$$

Applying Nernst Equation

$$E_{oP} = E^{\circ}_{Zn/Zn^{2+}} - \frac{RT}{nF}\ell n\left[Zn^{2+}_{(aq.)}\right]^1$$

With time

$$E^{\circ}_{Zn/Zn^{+2}} = \text{constant}$$

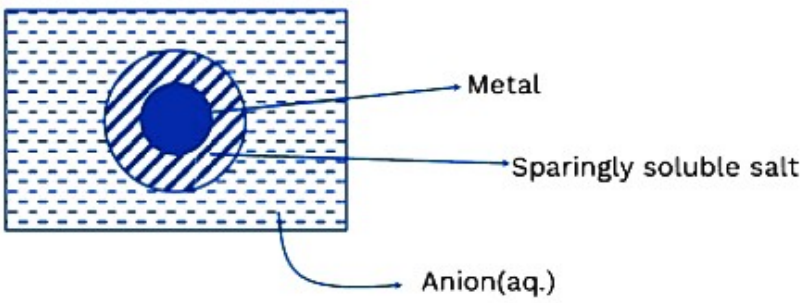
$$\frac{RT}{nF} = \text{constant}$$

Hence E_{OP} increase if $\left[Zn^{+2}\right]$ decreases and vice versa.

A constant Potential can be achieved using Metal-insoluble salt half-cell.

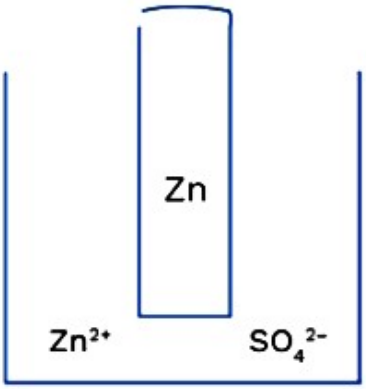
In this type of half-cell, a Metal electrode is first coated with saturated paste of its sparingly soluble salt & then is placed in aq. solution containing anion, of sparingly soluble salt.

Salt $\rightleftharpoons$ Metal cation (aq) + anion (aq) (Already done in ionic equilibrium)



Soluble salt of Metal (coated)

$ZnSO_4$ is coated. It is Sparingly soluble salt of zinc.



Example :

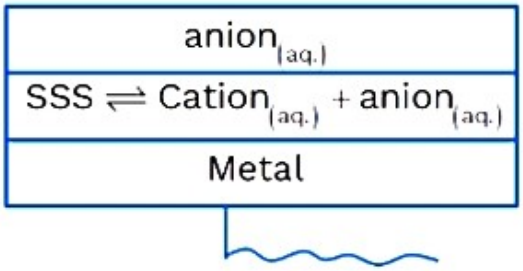
Metal	Sparingly soluble salt	Anion
Ag	AgCl AgBr AgI	Cl^-

Points to remember

For silver Metal except for $AgNO_3$ and AgF, mostly other salts of silver are sparingly soluble.

Working :

Three layer system created on electrode.



As anode

∵Oxidation

i.e., **Metal ⟶ Cation**

Since, we have sparingly soluble salt in equilibrium with its ions in the solution if concentration of Cation increases. Applying Le-Chatelier's Principle equilibrium shifts backward which maintains the equilibrium and concentration of cation in the solution.

Hence for a cell ⇒ [Cation] = Constant

Anode Representation

$Metal(s)|SSS_{(s)}|anion_{(aq.)}$

As Cathode

∵Reduction

i.e., **Cation ⟶ Metal**

Since, we have sparingly soluble salt in equilibrium with its ions in the solution if concentration of Cation decreases. Applying Le-Chatelier's Principle equilibrium shifts forward which maintains the equilibrium and concentration of cation in the solution.

Hence for a cell ⇒[Cation]=Constant

Cathode Representation

$anion_{(aq.)}|SSS_{(s)}|Metal_{(s)}$

Example :

$Ag_{(s)}\left|AgCl_{(s)}\right|Cl^-_{(ac.)}$

i.e. anode ⇒ oxidation

$$Ag_{(s)} \longrightarrow Ag^+(aq) + e^-$$

$$Cl^-_{(ac.)} + Ag(s) \rightleftharpoons AgCl_{(s)} + e^-$$

Applying Nernst equation.

$$E_{oP} = E^\circ_{Ag/AgCl/Cl^-} - \frac{RT}{nF}\ell n\frac{1}{\left[Cl^-\right]}$$

Example :

$Cl^-_{(aq.)}\left|PbCl_{2(s)}\right|Pb(s)$

$\because$ Cathode ⇒Reduction

i.e.

$$2e^- + \overset{+2}{Pb}Cl_2(s) \rightleftharpoons \overset{0}{Pb}(s) + 2Cl^-_{(aq.)}$$

$$E_{RP} = E^\circ_{Cl|PbCl_2|Pb} - \frac{RT}{nF}\ell n\left[Cl^-\right]^2$$

Example :

$$Pt_{(s)}\left|\overset{0}{Hg}_{(\ell)}\right|\overset{+1}{Hg_2}Cl_{2(s)}\left|Cl^-_{(aq.)}\right.$$

"Calomel electrode" as anode⇒Oxidation

i.e. $2Cl^-_{(aq.)} + 2Hg_{(\ell)} \rightleftharpoons Hg_2Cl_{2(s)} + 2e^-$

$$E_{op} = E^\circ_{op} - \frac{RT}{nF}\ell n\frac{1}{\left[Cl^-\right]^2}$$

Points to remember

Above is a special electrode of mercury known as Calomel electrode. In N(Normal)-Calomel electrode, $\left[Cl^-\right] = 1M$.

Calculation of electrode potential of Metal-insoluble salt half cell

This can be understood by taking an example of $Ag(s)\left|AgCl(s)\right|Cl^-_{(aq.)}$ half cell (anode)

Ionization: $Ag(s) \rightleftharpoons Ag^+_{(aq.)} + e^-; \Delta G^\circ_1$

Precipitation: $Ag^{+}_{(aq.)} + Cl^{-}_{(aq.)} \rightleftharpoons AgCl_{(s)};\ \Delta G^{\circ}_{2}$

Half cell reaction: $Ag^{+}Cl^{-}_{(aq.)} \rightleftharpoons AgCl_{(s)} + e^{-};\ \Delta G^{\circ}$

So, $\Delta G^{\circ} = \Delta G^{\circ}_{1} + \Delta G^{\circ}_{2}$

$$\left(-n.F.E^{\circ}\right) = \left(-n.f\ E^{\circ}_{Ag|Ag^{+}}\right) + \left(-RT\ell n\frac{1}{K_{sp}}\right)$$

$$E^{\circ} = E^{\circ}_{Ag|Ag^{+}} + \frac{RT}{nF}\ell n\frac{1}{K_{sp}}$$

So,

$$\left(E^{\circ}_{OP}\right)_{\substack{Metal\\SSS}} = \left(E^{\circ}_{OP}\right)_{\substack{Metal\\ion}} - \frac{RT}{nF}\ell n\ K_{sp}$$

Q24 **A silver dipped in 0.1 M HCl solution saturated with AgCl develops oxidation potential of −0.25 V. If $E^{\circ}_{Ag/Ag^{+}} = -0.799V$, then K_{sp} of AgCl in pure water will be**

(A) 2.95×10^{-11} **(B)** 5.1×10^{-11} **(C)** 3.95×10^{-11} **(D)** 1.95×10^{-11}

Sol. **(B)**

Half cell Reaction $Ag(s) + Cl^{-}_{(ag)} \rightleftharpoons AgCl_{(s)} + e^{-}$

$$E_{op} = E^{\circ}_{op} - \frac{0.059}{n}\log\frac{1}{\left[Cl^{-}\right]}$$ Since options close, no approximation use

$$-0.25 = E^{\circ}_{op} - \frac{0.59}{1}\log\frac{1}{0.1} \Rightarrow E^{\circ}_{op} = -0.191V$$

$$\text{Now}\left(E^{\circ}_{op}\right)\underset{SSS}{Metal} = \left(E^{\circ}_{op}\right)\underset{ion}{Metal} = \frac{0.059}{n}\log\ k_{sp}$$

$$-0.191V = -0.799 - \frac{0.059}{11}\log k_{sp}$$

$$0.799 - 0.19 = -0.059\log\ Ksp$$

$$-\left(\frac{0.608}{0.059}\right) = \log Ksp$$

$$-10.31 = \log\ Ksp$$

Taking anti log $Ksp = 5.1\times10^{-11}$

Q25 **In the cell $Pt(s)|H_2(g,1bar)|HCl(aq)|AgCl(s)|Ag(s)|$ the cell potential is 0.92 V when a 10^{-6} molal HCl solution is used. The standard electrode potential of $(AgCl/Ag,Cl^-)$ electrode is: $\left(\text{Given}: \frac{2.303RT}{F} = 0.06V \text{ at } 298K\right)$**

Sol. Half cell Reaction

At anode : $\frac{1}{2}H_2(g) \longrightarrow H^+(aq.) + e^-$

At cathode : $AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq.) + H^+(aq.)$

We know :

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ Let $E^\circ_{Ag^+/Ag} = x$

$= x - 0$

Standard Hydrogen potential = Zero

Applying Nernst Equation

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF}\log\theta$$

$$E_{cell} = E^\circ_{cell} - 0.06\log\left[Cl^-\right]\left[H^+\right]$$

$$0.92 = x - \frac{0.06}{1}\log\left\{10^{-6}\times 10^{-6}\right\}$$

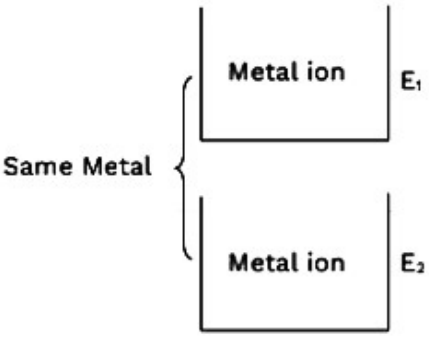
$x = -0.20$ volt

Points to remember

A "metal-ion half cell" and a "metal-sss half cell" produce same potential if [cation] in both are same.

Alternative method to solve questions on metal / metal sparingly soluble salt electrode

i.e.



If [cation] in both is same
$\Rightarrow E_1 = E_2$

Let for metal-sss half cell $Ag(s)|AgCl(s)|Cl^-_{(aq.)}$

$$Ag(s) + Cl^-_{(aq.)} \rightleftharpoons AgCl(s) + e^-$$

$$E = E^\circ - \frac{RT}{nF}\ln\frac{1}{[Cl^-]}$$

For sparingly soluble salt

$$AgCl(aq.) \rightleftharpoons Ag^+_{(aq.)} + Cl^-(aq.)$$

At equilibrium $K_{sp} = [Ag^+][Cl^-]$

$$E = \left(E^\circ_{Ag/Ag^+} - \frac{RT}{nF}\ln K_{sp}\right) - \frac{RT}{nF}\ln\frac{1}{[Cl^-]}$$

$$E = E^\circ_{Ag/Ag^+} - \frac{RT}{nF}\ln\frac{K_{sp}}{[Cl^-]}$$

For AgCl : $\frac{K_{sp}}{[Cl^-]} = [Ag^+]$

$$(E)_{\underset{sss}{metal}} = E^\circ_{Ag/Ag^+} - \frac{RT}{nF}\ln[Ag^+]$$

Metal-ion half cell: $Ag(s)/Ag^+_{(aq.)}$

HCR : $Ag(s) \rightleftharpoons Ag^+_{(aq.)} + e^-$

$$(E)_{\underset{ion}{metal}} = E^\circ_{Ag/Ag^+} - \frac{RT}{nF}\ln(Ag^+)$$

So

If $[Ag^+]$ in both are same

$$\Rightarrow (E)_{\underset{sss}{metal}} = (E)_{\underset{ion}{metal}}$$

Q26 **A silver wire dipped in 0.1 MHCL solution saturated with AgCl develops oxidation potential of −0.25 V. If $Ag^\circ_{Ag/Ag^+} = -0.799V$, the K_{sp} of AgCl in pure water will be**

(A) 2.95×10^{-11} **(B) 5.1×10^{-11}** **(C) 3.95×10^{-11}** **(D) 1.95×10^{-11}**

Sol. **(B)**

$Ag/AgCl/Cl^-$ $Ag(x) \mid Ag^+(aq).$

from k_{sp} of AgCl

$k_{sp} = [Ag^+]\,[Cl^-]^1$

$\Rightarrow \left[Ag^+\right] = \dfrac{k_{sp}}{\left[Cl^-\right]}$

$Ag(s) \rightleftharpoons Ag^+(aq) + e^-$

$E = E_{Ag/Ag^+} - \dfrac{0.059}{n}\log[Ag^+]$

$-0.25 = (0.799) - \dfrac{0.059}{1}\log\dfrac{k_{sp}}{0.1}$

$Ag(s) \mid AgCl(s) \mid Cl^-(aq.) \parallel Br^-(eq.) \mid AgBr(s) \mid Ag(s)$

$\Rightarrow Ag(s) \mid Ag^+(aq) \parallel Ag^+(aq.) \mid Ag(s)$

$E = 0 - \dfrac{0.059}{1}\log\dfrac{(k_1 / c_1)^1}{(k_2 / c_2)^1}$

Q27 **For the cell: $\overbrace{Zn(s) \mid Zn^{2+}(eq.)} \mid \overbrace{Cl^-(eq) \mid PbCl_2(s) \mid Pb(s)}$,Calculate the e.m.f of the cell.**

0.01 m **0.1 m** **$k_{sp} = 10^{-12}$**

Given: $E^\circ_{Zn^{2+}|Zn} = 0.76V$; $E^\circ_{Pb^{2+}|Pb} = 0.11V$

Sol. Representation of cell

$$\underbrace{Zn(s)\left|Zn^{2+}(eq.)\right|}_{\text{anode}}\ \underbrace{Cl^-(eq.)\left|PbCl_2(s)\right|Pb(s)}_{\text{cathode}}$$

$\Rightarrow Zn(s)\left|Zn^{2-1}_{(aq.)}\right|\left|Pb^{2+}_{(aq.)}\right|Pb(s)$

$PbCl_2(s) \rightleftharpoons Pb^{2+}_{(eq.)} + 2Cl^-(eq.)$

$K_{sp} = [Pb^{2+}]\,[Cl^-]^2$

$\Rightarrow [Pb^{2+}] = \dfrac{10^{-12}}{(0.1)^2} = 10^{-10}$

$\Rightarrow Zn(s) \mid Zn^{2-1}(aq.) \parallel Pb^{2+}(aq) \mid Pb(s)$

0.01 M from k_{sp}

Nett cell reaction

$Zn(s) + Pb^{+2}(aq) \rightleftharpoons Zn^{-2}(aq.) + Pb(s)$

$$E = E^\circ - \frac{0.054}{n}\log\frac{[Zn^{+2}]}{[Pb^{+1}]}$$

$$= (0.6) - \frac{0.06}{2}\log\frac{0.01}{10^{-10}} = 0.36V$$

Thermodynamics of Cell Potential

We know that :

$\Delta G^0 = -nFE^0_{cell}$
$\Delta G = -nFE_{cell}$
$G = H - TS$
From thermodynamics $H = E + PV$ enthalpy function.
Substituting $G = E + PV - TS$
By partial differentiation

$$\partial G = \partial E + P\partial V + V\partial P - T\partial S - S\partial T \qquad ...(i)$$

according to **1st law** of thermodynamics
$E = q + W$ $\qquad dW = -PdV$

$$\partial E = \partial q - PdV$$

$$\partial q = \partial E + P\partial V \qquad ...(ii)$$

according to **2nd law**

$$ds = \frac{dq}{T} = dq = Tds \qquad ...(iii)$$

From (i), (ii) and (iii)

$$dG = V\partial P - S\partial T$$

at constant pressure, which is actually the condition for all normal cell reaction.

$$dG = -S\partial T$$

$$S = \frac{-\partial G}{\partial T}$$

$$\Delta S = -\frac{\partial(\Delta G)}{\partial T}$$

$$\Delta G = -nFE_{cell}$$

$$\Delta S = +nF\frac{\partial E_{cell}}{\partial T}$$

$\frac{\partial E_{cell}}{\partial T}$ = temperature coefficient of cell reaction.
$\Delta G = \Delta H - T\Delta S$

$$\Delta H = -nF\,E_{cell} + nFT\,\frac{\partial E_{cell}}{\partial T}$$

Kirchoff's equation

$$\Delta Cp = \frac{\partial}{\partial T}(\Delta H)$$

Cp (of products) – Cp (of reactants)

$$= \frac{-nF\partial E_{cell}}{\partial T} + nFT\frac{\partial^2 E_{cell}}{\partial T^2} + \frac{-nF\partial E_{cell}}{\partial T}$$

$$\Delta Cp = nFT\frac{\partial^2 E_{cell}}{\partial T^2}$$

ELECTROLYSIS

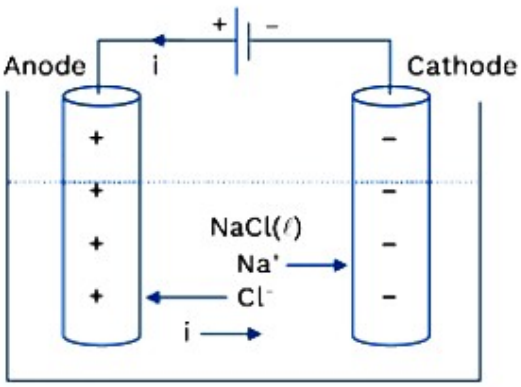
When current from an external source is passed into either **molten electrolyte or aqueous electron** solution then ions present in the solution move toward opposite electrodes and get discharged on their surface. This phenomenon is called electrolysis.

The cell used is called electrolytic cell. Electrolysis is generally used for purification of metals in metallurgy.

Electrolysis of molten NaCl

This discussion will give us better understanding of electrolysis. When molten NaCl is electrolysed, it dissociates into ions. Na^+ moves towards cathode and Cl^- moves towards anode. Current flows from anode to cathode.

Movement of ions between electrodes.



Anode

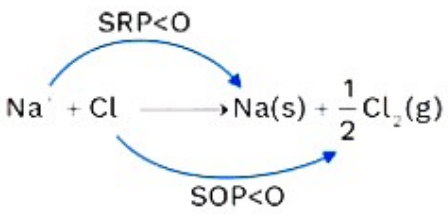
$$Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + 2e^-$$

i.e. Oxidation and standard oxidation potential (S.R.P)<0

Cathode

$Na^+(aq) + e^- \rightarrow Na(s)$

i.e reduction and standard reduction potential (S.R.P)< 0



Reversible sign not used, because chlorine will escape. According to Le-Chatelier, more probability of reaction to move in forward direction.

Now,

From Nernst equation : $E°_{cell} = E°_{OP} + E°_{RP}$

$\Rightarrow \quad E_{cell} < 0 \quad \Rightarrow \quad \Delta G > 0$

This indicates that the conversion of above ions into their elemental state is a non-spontaneous process as $\Delta G > 0$. Hence, we can say that electrolysis is carried out to make non spontaneous redox reactions feasible.

Above non-spontaneous redox reaction is taking place by means of external work done on the electrolytic cell.

Points to remember

- The solution shows conductivity due to movement of ions furnished by the electrolyte.

FARADAY'S LAWS OF ELECTROLYSIS

Faraday's First Law of electrolysis:

During electrolysis, the mass of a substance produced (or consumed) at an electrode is directly proportional to the charge utilized in the process.

Mathematically,

i.e. $w \alpha Q$

$\Rightarrow \quad w = Z.Q$

$= z. i. t$

Where,

w = mass of substance (in gram)

Q = charge utilized (Coulomb)

i = constant current (ampere)

t = time (in seconds)

Z = Electrochemical equivalent of substance.

Defining Electro chemical equivalent of substance (z) mathematically:

If Q = 1 coulomb charge

$\Rightarrow w = z$

i.e. Z is mass of substance produced when 1C charge is utilized.

Since,

Charge of 1 mole electron = 96500 Coulomb

Also, n- factor. of electron = 1

$\Rightarrow$ 96500 Coulomb = 1 equivalent of e^- So, for QC charge

$$= (\text{Equivalent})_{e^-} = \frac{Q(C)}{96500} = \frac{i \times t}{96500}$$

From law of equivalence:

$$\boxed{(\text{Equivalents})_{\text{substance}} = (\text{Equivalents})e^-}$$

$$\frac{W_{\text{substance}}}{E_{\text{substance}}} = \frac{Q}{96500}$$

$$\Rightarrow W_{sub} = \frac{E_{sub}}{96500} Q$$

On comparing with faraday's law

$$z = \frac{E}{96500}$$

For a substance $E = \frac{\text{Molar mass}}{\text{N} - \text{factor}}$

N-factor:

Net increase or decrease in Oxidation Number per Molecule

Example: $\overset{+1}{Na^+} \longrightarrow \overset{0}{Na}$

$$(\text{N.factor})_{Na^+} = 1$$

$$(\text{N.factor})_{Na} = 1$$

Example: $\overset{+1}{H^+} \longrightarrow \overset{0}{H_2}$

$$(E)_{H^+} = \frac{1}{1}$$

$$(E)_{H_2} = \frac{2}{2}$$

Example: $Mg^{2+} \longrightarrow \overset{0}{Mg}$

$$(E)_{Mg^{2+}} = \frac{24}{2}$$

$$(E)_{Mg} = \frac{24}{2}$$

Points to remember

- For electrolysis

$$(Eq)_{sub} = (Eq)_{e^-}$$

$$= \frac{w}{E} \qquad = \frac{Q}{96500}$$

$$= \frac{W}{M_w} \times N - fac \qquad = \frac{i \times t}{96500}$$

$$= n \times N.fac$$

$$= M \times V \times N.fac$$

$$= N \times V$$

Faraday's second law of electrolysis:

When same amount of charge is passed in different electrolytic cells, then at each electrode of every cell, substances are produced/ consumed in **equal no. of equivalents.**

$(Eq)_{sub} = (Eq)_{e^-}$

If same charge $\Rightarrow$ same $(Eq)_{e^-}$

i.e., $(Eq)_1 = (Eq)_2 = (Eq)_3 \ldots = (Eq)_{e^-}$

also in same cell

$(\text{Equivalents})_{anode} = (\text{Equivalents})_{cathode}$

Q28 **One gm metal M^{+2} was discharged by the passage of 1.81×10^{22} electrons. What is atomic weight of metal?**

(A) 33.35 **(B) 133.4** **(C) 66.7** **(D) 55**

Sol. **(C)**

$M^{2+} \longrightarrow M$

$(Eq)_M = (Eq)_{e^-}$

$\frac{w}{M_w} \times \text{N.factor} = (n)_{e^-} \times \text{N.factor}$

$\frac{1}{M} \times 2 = \left(\frac{1.8\times10^{22}}{6\times10^{23}}\right) \times 1$

$\frac{1}{M} \times 2 = \frac{3}{100}$

$M = \frac{200}{3} = 66.67\ g/mol$

Alternatively, from Farraday's Law

$w_M = z.Q$

$1 = \frac{M/2}{96500}\left(1.81\times10^{22} \times 1.6\times10^{-19} C\right)$

Q29 **Two Faraday of electricity is passed through a solution of $CuSO_4$. The mass of copper deposited at the cathode is : (at. Mass of Cu = 63.5 amu)**

[JEE Main 2015]

(A) 2 g **(B) 127 g** **(C) 0 g** **(D) 63.5 g**

Sol. **(D)**

According to Faraday's Law

$Cu^{2+} \longrightarrow Cu$

$(Eq)_{cu} = (Eq)_{e^-}$ (n-factor of Cu = 2)

$\frac{w}{63.5} \times 2 = \frac{2F}{F}$

$w = 63.5g$

Q30 **One mole of electron passes through each of the solution of $AgNO_3$, $CuSO_4$ and $AlCl_3$ then Ag, Cu and Al are deposited at cathode. The molar ratio of Ag, Cu and Al deposited are**

(A) 1 : 1 : 1 **(B) 6 : 3 : 2** **(C) 6 : 3 : 1** **(D) 1 : 3 : 6**

Sol. **(B)**

According to Faraday's second law

$Ag^{+} \longrightarrow Ag\,(s)$

$Cu^{2+} \longrightarrow Cu\,(s)$

$Al^{+3} \longrightarrow Al\,(s)$

Same Charge

Hence,

$(Eq)_{Ag} = (Eq)_{cu} = (Eq)_{Al} = (Eq)_{e^-}$ (equivalents = moles X n-factor)

$x \times 1 = y \times 2 = z \times 3 = 1 \times 1$

$\Rightarrow x = 1$ Mole

$y = \frac{1}{2}$ Mole

$z = \frac{1}{3}$ Mole

$\Rightarrow x : y : z = 6 : 3 : 2$

Current efficiency (η)

$$\eta = \frac{\text{charge utilised}}{\text{charge supplied}} \times 100$$

$$= \frac{(i)\,\text{current utilised}}{(i)\,\text{current supplied}} \times 100$$

Since electrolysis takes place due to **utilised charge**

i.e. according of Faraday's Law

$$w_{sub} = z(Q)\text{utilised}$$

Or

$$= z(i)\text{ utilized} \times \text{time}$$

Where $(i)\,\text{utilised} = \frac{\eta}{100} \times \text{supplised}$

Note:

If not mentioned in the question, efficiency is to be assumed as 100%.

Electrolysis of aqueous solution

In aqueous solutions, there are following three substances present:

(i) Cation of electrolyte

(ii) Anion of electrolyte

(iii) Water

When electricity is passed in such solution, then the preferential of ion at an electrode with respect to water will depend upon its discharge potential $(E_{op}$ or $E_{Rp})$

Points to remember

If concentrations of all ions & partial pressure of all gases involved in half Cell reactions are unity, then E can be taken as the criterion for preferential discharge.

At anode

Anion (aq.) & $H_2O(\ell)$

If $(E_{OP})_{anion} > (E_{OP})_{H_2O}$

$\Rightarrow$ Oxidization of anion will take place & vice-versa

At cathode

Cation (aq.) & H_2O

If $(E_{RP})_{cation} > (E_{RP})_{H_2O}$

$\Rightarrow$ Reduction of cation take place & vice-versa

For electrolysis of IM aqueous electrolyte using inert electrodes

1)	**Reduction tendency of cations**	Cations of 1st, 2nd and 13th group $< H_2O <$ Rest cations i.e., between Na^+ & H_2O $\Rightarrow$ Reduction of H_2O but, between Fe^{2+} & H_2O $\Rightarrow$ Reduction of Fe^{2+}
2)	**Oxidation tendency of anions**	$I^-, Br^-, Cl^- > H_2O >$ Rest anions For example between Cl^- & $H_2O \Rightarrow$ Oxide of Cl^- But, between NO_3^- & $H_2O \Rightarrow$ Oxide of H_2O
3)	**Oxidation of H_2O**	Half Cell Reaction $2\overset{+1}{H_2}\overset{-2}{O}(\ell) \longrightarrow O_2(g) + 4H_{(aq.)} + 4e^-$ $E^\circ_{H_2O/O_2} = (-1.23 \text{ volt})$ As per chemical kinetics, the oxidation of H_2O is a slow process, as it requires large activation energy. The reaction will be slow. Hence, to speed up the above reaction, energy is provided by some extra voltage from battery which is known as over voltage or over potential. So, Effectively the discharge potential for oxidation of $H_2O = -(1.23 + x)$Volt Where x is over potential The value of x depends upon the type of electrode used. e.g. for platinised Pt electrode, x = 0.25 for smooth Pt electrode, x = 0.45 $\rightarrow$Due to oxidation of H_2O, acidic nature increases.
4).	**Reduction of H_2O**	**Half Cell Reaction:** $2H_2O_{(\ell)} + 2e^- \longrightarrow H_{2(g)} \uparrow + 2OH^-_{(aq.)}$ $\rightarrow$ Due to reduction of H_2O basic nature increases. $E^\circ_{H_2O/H_2} = (-0.83V)$

Some examples given below will help us understand the electrolysis process of aqueous solution better.

Electrolysis of 1 M HCl (aq) solution

Anode:

Among $Cl^-_{(aq)}$ & H_2O, oxidation tendency of chloride ion is higher.
So, half cell reaction

$$2Cl^-_{(aq)} \longrightarrow Cl_2 + 2e^-$$

Cathode:

Among $H^+_{(aq)}$ & $H_2O(\ell)$

Reduction tendency of H^+ is higher
So, Half Cell Reaction

$$2H^+ + 2e^- \rightarrow H_2(g)$$

Net Cell Reaction

$$2H^+ + 2Cl^-(aq.) \rightarrow H_2(g) + Cl_2(g)$$

$$\underbrace{(Eq)_{H^+} = (Eq)_{Cl^-}}_{\text{Consumed}} = \underbrace{(Eq)_{H_2} = (Eq)_{Cl_2}}_{\text{produced}}$$

***If $[Cl^-] < 1$ M , then**

$$(E_{op})_{Cl^-} < (E_{op})_{H_2O}$$

$\Rightarrow$ oxidation of H_2O

$\Rightarrow$ O_2 will be evolved at anode

In short:

Electrolysis of aqueous electrolyse (1<M)

In Nernst equation $E = E^\circ - \frac{0.059}{n}^{1} \log k(\text{for 1 M})$

Hence $E_{R.P.} = E^\circ_{R.P.}$

For water $\rightarrow$
Oxidation of H_2O

$$\overset{+1\ -2}{H_2O} \longrightarrow \overset{0}{O_2}$$

Reduction of H_2O

$$\overset{+1\ -2}{H_2O} \longrightarrow \overset{0}{H_2}$$

Electrolysis of 1 M NaCl(aq) solution

Anode

Among $Cl^-_{(aq)}$ & H_2O

Chloride ion has more oxidation potential.

Half cell reaction

$$2Cl^-_{(aq)} \longrightarrow Cl_2 - 2e^-$$

Cathode

Among $Na^+_{(aq)}$ & $H_2O(\ell)$

Water has more reduction potential.

Half cell reaction

$$2H_2O_{(\ell)} + 2e^- \longrightarrow H_{2(g)} + 2OH^-_{(ag.)}$$

Nett Cell Reaction

$$2H_2O(\ell) + 2Cl^-_{(aq.)} \longrightarrow H_2(s) + Cl_2(s) + 2OH^-_{(aq)}$$

$$\underbrace{(Eq)_{H_2O} = (Eq)_{Cl^-}}_{\text{Consumed}} = \underbrace{(Eq)_{H_2} = (Eq)_{Cl_2} = (Eq)_{OH^-} = (Eq)_C}_{\text{produced}}$$

If dilute $NaCl_{(aq.)}$ $\Rightarrow$ Oxidation of H_2O & Reduction of H_2O

i.e., [NaCl] < 1 M $\Rightarrow$ Nett Cell Reaction: $2H_2O(\ell) \rightleftharpoons 2H_{2(g)} + O_2(g)$

Electrolysis of 1 M H_2SO_4(aq) solution

Anode

Among $SO^{2-}_{4(aq)}$ & $H_2O(\ell)$

Sulphate ion has lower oxidation potential.

$$2H_2O(\ell) \longrightarrow O_2(s) + 4H^+_{(aq.)} + 4e^- \times 1$$

Cathode

Among $H^+_{(aq.)}$ & $H_2O(\ell)$

Water has more reduction potential.

$$2H_2O(\ell) + 2e^- \longrightarrow H_2(s) + 2OH^-(aq.) \times 2$$

Nett Cell Reaction

$$2H_2O(\ell) \longrightarrow 2H_2(s) + O(g) + \cancel{\left(4H^+(aq.) + 4OH^-(aq.)\right)}$$

$$\underbrace{(Eq)_{H_2O}}_{\text{consumed}} = \underbrace{(Eq)_{H_2O} = (Eq)_{O_2}}_{\text{consumed}} = (Eq)_{e^-}$$

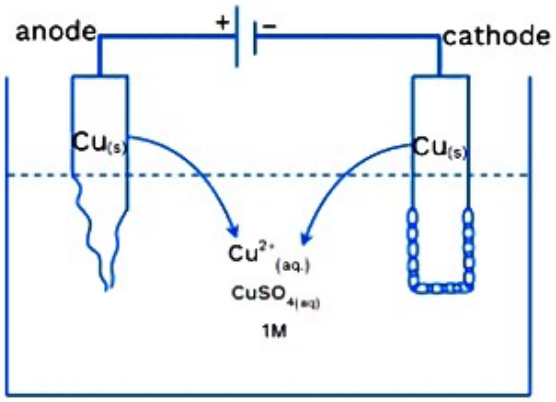
Electrolysis of 1M $CuSO_4$(aq) solution

Case-I: Using inert electrodes

Anode	Among $SO_{4(aq.)}^{2-}$ & $H_2O(\ell)$ Sulphate ion has lower oxidation potential $2H_2O(\ell) \longrightarrow O_2(s) + 4H^+_{(aq.)} + 4e^- \times 1$
Cathode	Among $Cu^{2+}_{(ac.)}$ & $H_2O(\ell)$ water has lower oxidation potential Half Cell Reaction $Cu^{2+}_{(ac)} + 2e^- \longrightarrow Cu(s) \times 2$
Nett cell Reaction	$2Cu^{2+}_{(aq.)} + 2H_2O(\ell) \longrightarrow 2Cu(s) + O_2(s) + 4H^+_{(aq)}$ $\underbrace{(Eq)_{Cu^{2+}} = (Eq)_{H_2O}}_{\text{consumed}} = \underbrace{(Eq)_{Cu} = (Eq)_{O_2} = (Eq)_{H^+}}_{\text{Produced}} = (Eq)_{e^-}$

Case-II: Using copper electrodes (React-able or attackable electrodes).

Anode	$SO_{4(aq)}^{2-}, \underset{-(1.23-x)}{H_2O(\ell)}$ .and $\underset{(-0.34V)}{Cu(s)}$ Three options are available for oxidation. On comparing SO_4^{-2} and water, water gets oxidised. Among water and Cu, Cu has more tendency to show oxidation. Hence, final reaction at anode $\Rightarrow Cu(s) \rightarrow Cu^{2+}_{(aq.)} + 2e^-$
Cathode	Among $Cu^{2+}_{(aq.)}$ and $H_2O(\ell)$ Cu^{2+} will get reduces. Hence, reaction at cathode will be $\Rightarrow Cu^{2+}_{(aq.)} + 2e^- \longrightarrow Cu(s)$
Nett Cell reaction	$\underset{\text{anode (dissolution)}}{Cu(s)} \longrightarrow \underset{\text{Cathode (deposition)}}{Cu(s)}$



Points to remember

Size of anode decreases and size of cathode increases with the process. When attackable electrodes are used, concentration of solution (electrolyte) remains unchanged

Case-III: Using impure anode of copper

Anode

$\underset{x}{SO_4^{2-}}, \underset{-(1.23)}{H_2O}, \underset{(+0.49)}{Fe}, \underset{(0.76)}{Zn}, \underset{(-0.34)}{Cu}, \underset{(-0.8)}{Ag}, Au$

Zn and Fe have high oxidation potential and will oxidise initially. But since both metals are present in traces. Whatever amount present in anode will release in electrolyte

So First, $Zn(s) \longrightarrow Zn^{2+}_{(aq)} + 2e^-$

then $Fe(s) \longrightarrow Fe^{2+}_{(aq)} + 2e^-$

after zinc and iron metals are oxidised. Copper will start oxidation. Since, copper is present in high amount. The oxidation of copper will continue for longer time.

& then $Cu(s) \longrightarrow Cu^{2+}_{(aq)} - 2e^-$

All five metals were homogeneously mixed, and atoms were bonded with metallic bonding. So as oxidation takes place, the silver and gold ions will be free and will be collected at anode. This is known as anode mud.

Cathode

Ions $\underset{(-0.76v)}{Zn^{2+}_{(ac.)}}, \underset{(-0.44)}{Fe^{2+}_{(aq.)}}, \underset{(+0.34)}{Cu^{2+}_{(aq)}}, \underset{x}{H_2O(\ell)}$

Among all cations, only copper will reduce and deposit on cathode.

Only $Cu^{2+}_{(aq.)} + 2e^- \longrightarrow Cu(s)$

From Law of equivalence

Equivalents at anode = equivalents at cathode

So, $(Eq)_{anode} = (Eq)_{cathode}$

$\Rightarrow \left[(Eq)_{cu} + (Eq)_{Fe} + (Eq)_{zn}\right] = (Eq)_{Cu}$

Nett cell reaction

$Fe(s) + Zn(s) + Cu(s) \rightarrow Cu(s) + Fe^{2+}_{(aq.)} + Zn^{2+}_{(aq.)}$

Example :

Electrolysis of Molten NaCl using Hg Cathode

at cathode

In general $Na^{+} + e^{-} \longrightarrow Na_{(s)}$

If platinum electrode used , then sodium metal will deposit on it.

But for mercury electrode, metal diffuses in mercury and is known as amalgam.

& $Na_{(s)} + Hg(\ell) \longrightarrow \underset{(Amalgam)}{Na - Hg}$

Concentration term used for amalgam

$$\% \text{ of amalgam} = \frac{\text{wf. of Metal (gram)}}{\text{wf. of amalgam (gram)}} \times 100$$

i.e. x % M-Hg amalgam :

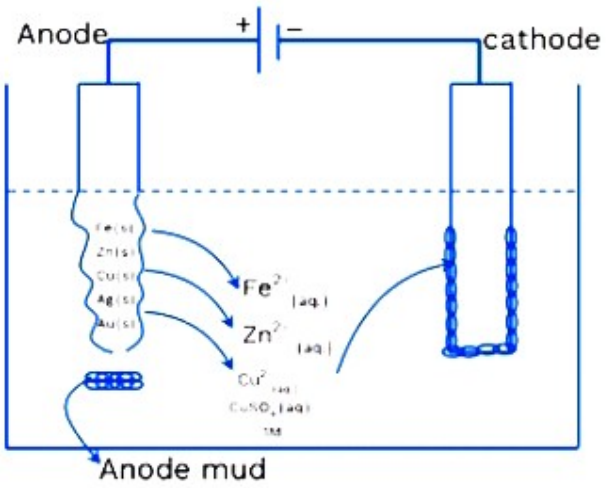
Let x gram metal in 100 gram amalgam (solute)

$\Rightarrow$ amount of Hg = (100 – x)gram (solvent)

Note :

For max amount of amalgam:

$(n)_{Metal} = (n)_{Hg}$



Points to remember

Used in electrorefining of metals, to get pure metals from their ore. Anode is made up of impure metal and cathode is made from pure metal.

Points to remember

During electrolysis, if Hg-cathode or Calomel cathode is used then upon reduction of metal-ion, amalgam is formed at cathode.

Amalgam → alloy of Hg with a Metal.

Q31 **Magnesium amalgam is prepared by electrolysis of a molten $MgCl_2$ using a mercury cathode. How long should a constant current of 9.65 A be passed in order to prepare 12% Mg-Hg amalgam on a cathode containing 8.8 gm Hg?**

Sol. Applying law of equivalence

$$(E_a)_{Mg} = (E_a)_{e^-} = \frac{9.65 \times t}{96500}$$

12g Mg is present in 100 g amalgam

$\Rightarrow$ 88 g Hg is present

$\because$ for 88g Hg $\longrightarrow$ 12g Mg

$\Rightarrow$ 8.8g Hg $\longrightarrow$ 1.2g Mg

$(Eq)_{Mg} = (Eq)_{e^-}$

$$\frac{1.2}{24} \times 2 = \frac{9.65 \times t}{96500}$$

t = 1000 sec

Electrolytic conductance (G)

The ability of a molten electrolyte or **aqueous electrolyte solution** to allow the passage of electricity through it

It is due to movement of ions furnished by the electrolyte

Matter transfer takes place.

A chemical reaction takes place.

It increases with increases in temperature

Metallic conductance (in physics) decreases with increase in temperature, but electrolytic conductance increases with increase in temperature.

Factors affecting Electrolytic conductance

(i) Solute-solute interaction:
For better conductance- the electrolyte should easily break in ions i.e intermolecular interaction between solute and solute particle should be weak. It relates with attraction between cation & anion

(ii) Solvent-Solvent interaction:
Ions moved inside solvent. There will be interaction between ions and water (solvent) particles. Ions will move in between gap of solvent particles.
For easy movement of ions, gaps between solvent particles should be high for better conductance,
It relates with viscosity of the solvent

(iii) Solute-Solvent interaction:
The interaction between solute and solvent should be low for better conductance.
It relates with solution or hydration of ions.

For good conductance, all above three attractions should be weak.
On increasing temperature, since all types of attractions become weaker hence, electrolyte conductance increases

Resistance

inverse of conductance.
Means the ability of a substance to oppose the passage of electricity through it.
Measured in ohm(Ω)

For an electrolytic cell

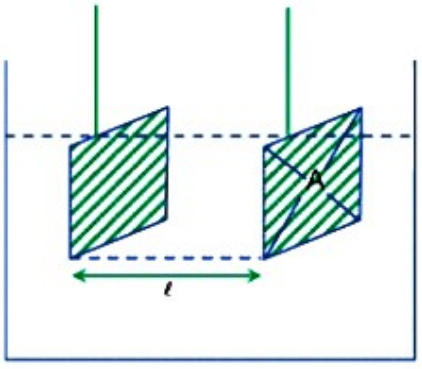
A → common interfacial area of electrodes (cm^2)
ℓ → distance between electrodes (cm)

$$\Rightarrow \boxed{\left(\frac{\ell}{A}\right) = G^* \text{cell constant}\,(cm^{-1})}$$

& $R \propto G^*$ where ρ → Resistivity/Specific resistance (Ω.cm)
$\Rightarrow R = \rho G^*$ G → Conductance of solution between electrodes

or $\frac{1}{\rho} = \frac{1}{R}.G^*$ Ω^{-1} or Mho or Siemens(s)

$\boxed{K = G.G^*}$ K → conductivity /Specific conductance/kappa ($\Omega^{-1}CM^{-1}$ or $S.CM^{-1}$)



Cell diagram

If $\left.\begin{matrix}\ell = 1cm \\ A = 1\ cm\end{matrix}\right\} G^* = 1cm^{-1}$

$\Rightarrow$ Volume of solution between electrons = 1 cm^3

$\because$ $K = G.G^*$

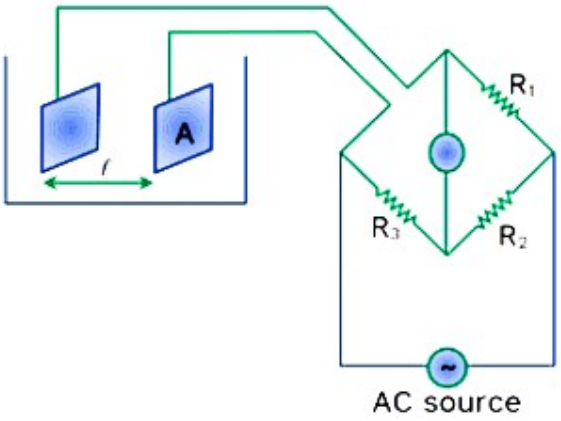
$\Rightarrow$ $K = G$

i.e. K is the conductance of $1cm^3$(or 1 m^3) of solution irrespective of moles of electrolyte

$\Rightarrow$ k → intensive property

Experimental determination of K & G

G* will be known from dimensions of electrodes. Experimentally, then resistance of solutions is determined using "Wheat stone bridge"



From the concepts of WB

$R_1.R_3 = R_2.R_{Solution}$

$\Rightarrow R_{solution} = \dfrac{R_1.R}{R_2}$

Now $\boxed{\dfrac{1}{R_{solution}} = G_{solution}}$

Also $G.G^* = K$

Note:

If direct current is used, then electrolytic decomposition takes place (electrolysis). Due to the composition of solution is changed. Hence, alternating current is used.

Molar conductance / Molar conductivity (Λ_M)

It is the conductance of a solution containing one mole of electrolyte irrespective of its volume

k → conductance of 1 cm^3 solution

Let take V cm^3 of solution

$K \times v = G_{solution}$

If the solution contains one mole electrolyte

$\Rightarrow \quad G_{solution} = \Lambda_M$

$\Rightarrow \quad \Lambda_M = k \,.\, v$

$\downarrow \downarrow$

$S.cm^{-1}\ cm^3.mol^{-1}$

So molarity of solution $= \dfrac{1 \text{ mole}}{\frac{v}{100}L}$

$M = \dfrac{100}{V} \Rightarrow V = \dfrac{100}{M}$

$\Rightarrow \quad \boxed{\Lambda_M = \dfrac{k \times 100}{M}}$ or $\boxed{\Lambda_M = \dfrac{k}{100.M}}$

Equivalent conductance / Equivalent conductivity (Λ_{eq})

The conductance of solution containing one gm. equivalent of electrolyte irrespective of volume

So $\boxed{\Lambda_{eq} = \frac{k \times 100}{N}}$

$\boxed{\Lambda_{eq} = \frac{k}{1000 \times N}}$

Note: For a solution

$\because$ $N = M \times N\text{-factor}$

$$\frac{k \times 1000}{\Lambda_{eq}} = \frac{k \times 1000}{\Lambda_M} \times N.fac$$

$\Rightarrow$ $\boxed{\Lambda_M = \Lambda_{eq} \times N - fac}$

Questions will be formula based

Q32 **Resistance of 0.2M solution of an electrolyte is 50Ω. The specific conductance of the solution is 1.4S m^{-1}. The resistance of 0.5 M solution of the same electrolyte is 280Ω. The molar conductivity of 0.5 M solution of the electrolyte in S m^2 mol^{-1} is** **[JEE Main 2014]**

(A) 5×10^{-3} **(B) 5×10^{3}** **(C) 5×10^{2}** **(D) 5×10^{-4}**

Sol. **(A)**

$G = \frac{1}{20}\Omega^{-1}$

$1.4 = \frac{1}{50} \times G^*$

$G^* = 70\ m^{-1}$

$k = \frac{1}{280} \times 70 = \frac{1}{4} S.m^{-1}$

Now $\Lambda_M = \frac{k}{1000 \times M}$

$= \frac{1/4}{1000 \times 1/2}$

$= 5 \times 10^{-4}\ S.M^2.\ Mol^{-1}$

G.G.*

$G^* = \frac{l}{A}$ $G = \frac{1}{R}$

$k = \frac{1}{\rho}$

$\Lambda_m = \frac{k \times 1000}{M}$

Points to remember

Four types of conductance

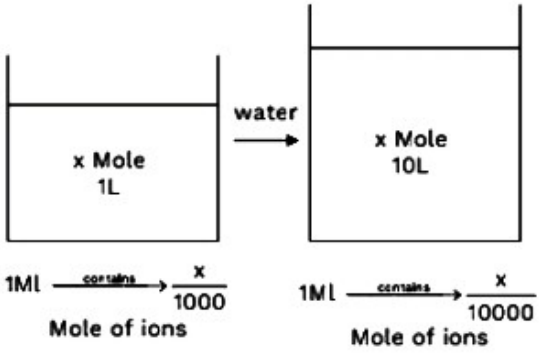
1. Conductance of solution (G)- extensive property
2. Specific conductance- conductance per unit volume (K) (symbol kappa)(intensive)
3. Molar conductance- conductance of solution with 1molar solution Λ_m (intensive)

Specific conductance- conductance of solution with 1 equivalent solution Λeq (intensive)

Variation Upon dilution :

For both strong and weak electrolyte, on dilution concentration decreases.

K(kappa) decreases because number of ions per unit volume decreases.



Example:

On diluting solution from 1L to 10 L number of ions present per ml also decreases.

$\wedge_m$ increase upon dilution because ions become more mobile.

Example:

On diluting 1 molar solution to 10 L.

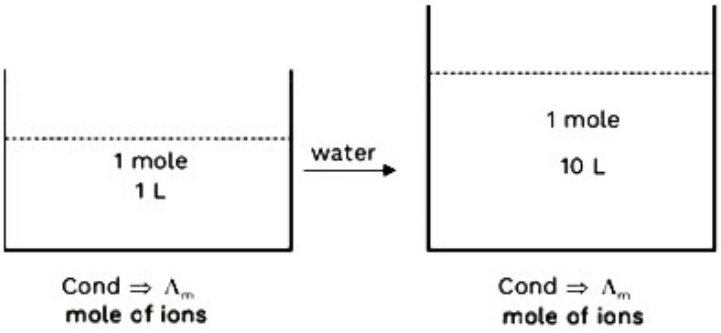
Moles of electrolyte in both solution = 1M

Conductance in both solution will be same Λ_m.

In the solution containing 1 mole particles in 1L, the movement of particles is difficult with respect to solution of volume 10 L.

Hence, conductance increases.

i.e on dilution conductance (G) increases i.e Λ_m increases the variation in Λ_m upon dilution ,depends upon the type of electrolyte as following :



Case-I:

For strong electrolytes
(i.e. complete ionisation)

Experimentally

$$\boxed{\Lambda_m = \Lambda_m^0 - A \cdot \sqrt{C}}$$

Where

C → Molarity of solution

Λ_m → Molar conductance at C concentration

Λ_m^0 → Limiting molar conductance

(i.e maximum. Λ_m)

$A \rightarrow (+)$ constant

So upon dilution,

If $\Lambda_m \rightarrow y$ – axis

$\sqrt{c} \rightarrow x$ – axis

$\Rightarrow \quad y = \Lambda_m^0 - A.x$

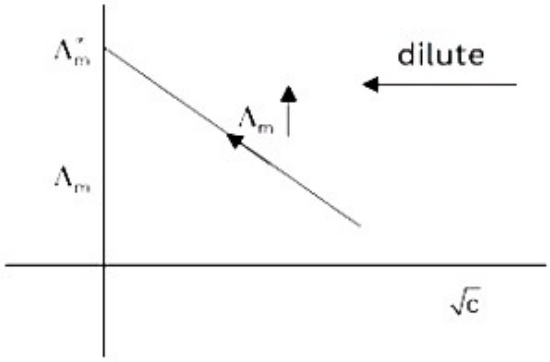
$\because \quad y = c + mx$

i.e. straight line with (–)ve slope and positive intercept.

Upon diluting, once Λ_m^0 is obtained, there will be no further increase in Λ_m° upon further dilution. Hence Λ_m^- can also be called as **molar conductance at infinite dilution** (Λ_m^∞).

Strong electrolyte	**Type for A**
NaCl	1-1 type
K_2SO_4	1-2 type
$MgCl_2$	2-1 type
$Ca_3(PO_4)_2$	2-3 type

Same type of strong electrolytes have same value of A and vice-versa.



Points to remember

For a given solvent at a particular temp. the value of constant 'A' depends upon the type of strong electrolyte which are distinguished on the basis of |charges| on cation and anion furnished by the electrolyte.

Case-II:

For weak electrolytes

i.e. Partial ionisation

So, upon dilution, Volume of solution increases.

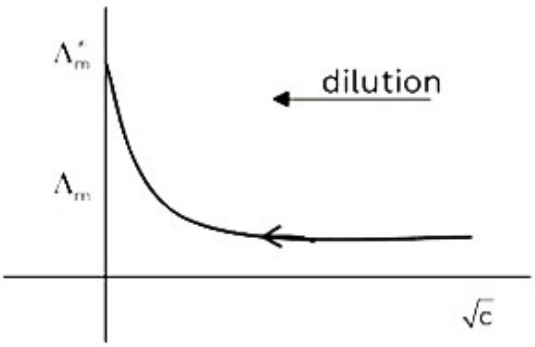
According to ostwalds law of dilution

$\Rightarrow$ number of ions increases.

$\Rightarrow \Lambda_m$ also increases but slowly

(V increases but increases in number of ions is in under root terms)

$\Rightarrow$ graph non linear



Dilution

Points to remember

- For weak electrolytes, Λ_m^0 cannot be determined practically but theoretically using Kohlrausch law.
- For a weak electrolyte at 'c' molar concentration.

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\infty}$$

$\Lambda_m \longrightarrow$ molar conductance. at c molar solution

$\Lambda_m^\infty \longrightarrow$ limiting molar conductance

KOHLRAUSCH LAW

At infinite dilution, where ions move freely, each ion gives its finite contribution to the **total molar conductance of electrolyte.** At infinite dilution molar conductance is maximum.

i.e. $\Lambda_m^\infty(\text{electrolyte}) = \Sigma(\text{contribution of ions})$

Example: $\Lambda_m^\infty(NaCl) = 1.\lambda_m^\infty(Na+) + 1.\lambda_m^\infty(Cl^-)$

where $\lambda_m^\infty \rightarrow$ molar ionic conductance at infinite dilutions.

$\Lambda_m^\infty(MgCl_2) = 1.\lambda_m^\infty(Mg^{2+}) + 2.\lambda_m^\infty(Cl^-)$

$\Lambda_m^\infty(AlCl_3) = 1.\lambda_m^\infty(Al^{3+}) + 3.\lambda_m^\infty(Cl^-)$

$\Lambda_m^\infty(Ba_3(PO_4)_2) = 3.\lambda_m^\infty(Ba^{2+}) + 2.\lambda_m^\infty(PO_4^{3-})$

i.e. In general

$$\boxed{\Lambda_m^\infty(A_xB_y) = x\,\lambda_m^\infty(A^{y+}) + y\,\lambda_m^\infty(B^x)}$$

***But

$$\boxed{\Lambda_{eq}^\infty(A_xB_y) = \lambda_{eq}^\infty(A^{y+}) + \lambda_{eq}^\infty(B^x)}$$

$$\Lambda_m = \Lambda_{eq} \times N - \text{factor}$$

So $\Lambda_{eq}^\infty(A_xB_y) \times (xy) = x\left[\lambda_{eq}^\infty(A^{y+})\,xy\right] + y\left[\lambda_{eq}^\infty(B^x)\,xy\right]$

$\Rightarrow \Lambda_{eq}^\infty(A_xB_y) = \lambda_{eq}^\infty(A^{y+}) + \lambda_{eq}^\infty(B^{x-})$

Q33 **The molar conductivities Λ^0_{NaOAc} and Λ^0_{HCl} at infinite dilution in water at 25°C are 91.0 and 426.5 S cm² / mol respectively. To calculate Λ^0_{HOAc} the additional value required is**

(A) Λ^0_{KCl} **(B)** Λ^0_{NaOH} **(C)** Λ^0_{NaCl} **(D)** $\Lambda^0_{H_2O}$

Sol. **(C)**

Given $\Lambda^{\circ}_{NaO\,Ac} = \Lambda^{\circ}_{Na^+} + \Lambda_{OAc^-}$ ... (i)

$\Lambda^{\circ}_{HCl} = \Lambda^{\circ}_{H^+} + \Lambda^{\circ}_{Cl^-}$ ... (ii)

To calculate $\Lambda^{c}_{HOAc} = \Lambda^{\circ}_{H^+} + \Lambda^{\circ}_{OAc^-}$ ... (iii)

Equations (i) – (ii) = (iii) - $\underbrace{\left\{\Lambda^{c}_{m}Na^+ + \Lambda^{c}_{m}Cl^-\right\}}_{\Lambda^{1}_{m}NaCl^-}$

Q34 **The equivalent conductance of two strong electrolytes at infinite dilution in H_2O (where ions move freely through a solution) at 25°C are given below:**
$\Lambda^{\circ}_{CH_3COONa} = 91.0\ S\ cm^2\ /\ equiv$; $\Lambda^{\circ}_{HCl} = 426.2\ S\ cm^2\ /\ equiv$

What additional information/quantity one needs to calculate Λ° of an aqueous solution of acetic acid?

(A) The limiting equivalent conductance of $H^+ (\lambda^{\circ}_{H^+})$

(B) Λ° of chloroacetic acid ($ClCH_2COOH$)

(C) Λ° of NaCl

(D) Λ° of CH_3COOK

Sol. **(C)**

Given $\Lambda^{\circ}_{CH_3COONa} = \Lambda^{\circ}_{CH_3COO^-} + \Lambda^{+}_{Na}$ ... (i)

$\Lambda^{\circ}_{HCl} = \Lambda^{\circ}_{H^+} + \Lambda^{\circ}_{Cl^-}$ ... (ii)

To calculate $\Lambda^{\circ}_{CH_3COOH} = \Lambda^{\circ}_{CH_3COO^-} + \Lambda^{\circ}_{H^+}$ ... (iii)

Equations (i) – (ii) = (iii) - $\underbrace{\left\{\Lambda^{c}_{Na^+} + \Lambda^{\circ}_{Cl^-}\right\}}_{\Lambda^{\circ}_{NaCl}}$

Q35 **The highest electrical conductivity of the following aqueous solutions is of**

(A) 0.1 M acetic acid **(B) 0.1 M chloroacetic acid**

(C) 0.1 M fluoroacetic acid **(D) 0.1 M difluoroacetic acid**

Sol. **(D)**

Since all are weak electrolytes, and same concemtration, the solution producing more ions will show higher conductivity

Q36 **The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120cm with an area of cross section of 1cm². The conductance of this solution was found to be 5×10⁻⁷S. The pH of the solution is 4. The value of limiting molar conductivity (Λ_m°) of this weak monobasic acid in aqueous solution is Z × 10²S cm⁻¹ mol⁻¹. The value of Z is:** **[JEE Advanced 2017]**

Sol. Given : $\left.\begin{matrix} G^* = \frac{120}{1}\text{cm}^{-1} \\ G = 5\times10^{-7}\text{S} \end{matrix}\right\} \Rightarrow k = G.G^* = 6\times10^{-5}\text{S.cm}^{-1}$

$$\Lambda_m = \frac{k\times1000}{M} = \frac{6\times10^{-3}\times1000}{1.5\times10^{-3}} = 40\ \text{S.cm}^2\ \text{mol}^{-1}$$

To calculate limiting molar conductivity (Λ_m°)

pH = 4

$\Rightarrow$ $[H^+] = 10^{-4}M = C\alpha$

$$\Rightarrow \alpha = \frac{1}{15} = \frac{\Lambda_m}{\Lambda_m^0}$$

$$\Lambda_m^0 = 15\times40 = 600\ \text{s.cm}^1\text{.mol}^{-1}$$

Q37 **The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.10M). If $\lambda_{X^-}^\circ \approx \lambda_{Y^-}^\circ$,the difference in their values, $pK_a(HX) - pK_a(HY)$, is (consider degree of ionisation of both acids to be <<1)** **[JEE Advance 2015]**

Sol. Given $\frac{\Lambda_m(HX)}{\Lambda_m(HY)} = \frac{1}{10}$ $\qquad \log Na(HX) = \log Na(HY) = -3$

$$\frac{\alpha_1}{\alpha_2} = \frac{\Lambda_m(HX)}{\Lambda_m^\circ(HX)} \Big/ \frac{\Lambda_m(HY)}{\Lambda_m^\circ(HY)} = \frac{1}{10} \qquad pKa(HX) - pKa(HY) = 3$$

$$\frac{Ka(HX)}{Ka(HY)} = \frac{C_1\alpha_1^2}{C_2\alpha_2^2} = \frac{0.01}{0.1}\times\left(\frac{1}{10}\right)^2 = \frac{1}{1000}$$

Q38 **The molar conductivity of a solution of a weak acid HX (0.01M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.10M). If $\lambda^0_{X^-} \approx \lambda^0_{Y^-}$, the difference in their pK_a value $pK_a(HX) - pK_a(HY)$, is (consider degree of ionisation of both acids to be <<1).**

Sol. $$\frac{\Lambda_m(HX)}{\Lambda_m(HY)} = \frac{1}{10}$$

$$\propto = \frac{\Lambda_m}{\Lambda_m^\infty} \quad \propto = \frac{\lambda_m}{\lambda_m^\infty}$$

$$\frac{\Lambda_n^0 HY}{\Lambda_m^0 HX} = \frac{\lambda^0_{mH^+} + \lambda^0_{nY^-}}{\lambda_m^- H^+ + \lambda_n^p X^-}$$

$$ka = c\alpha^2$$

$$\frac{ka(Hx)}{ka(Hy)} = \frac{c_1}{c_2}\left(\frac{\alpha_1}{\alpha_2}\right)^2$$

$$= \frac{0.01}{0.1}\left(\frac{\Lambda_m(HX)}{\Lambda_m(HY)}\right)^2 = \frac{1}{10}\left(\frac{1}{10}\right)^2 = 10^{-3}$$

$$\log k_a(Hx) - \log k_a(Hy) = -3$$

$$\Rightarrow pk_a(Hx) - pk_a(Hy) = 3$$

Conductance of saturated solution of Sparingly Soluble Solution :

Let AB

i.e. $AB_{(s)} \rightleftharpoons A^+_{(aq)} + B^+_{(aq)}$; K_{sp}

t = o	a	-	-
t_{eq}	a-s	s	s

s → solubility (mol/L)

$$\Lambda_m = \frac{k \times 1000}{s}$$

Also : upon dilution, reaction moves in forward direction & solubility remains constant

Hence : Λ_m also remains same

i.e. Λ_m^∞ (for Sparingly Soluble Salt) = Λ_m

Points to remember

$\because (G)_{sol^n} = (G)_{solute} + (G)_{solvent}$

$\because K = G.G^*$

$\Rightarrow (K)_{sol^n} = (K)_{solute} + (K)_{solvent}$

$\Rightarrow \boxed{(K)_{solute} = (K)_{sol^n} - (K)_{solvent}}$

If $(K)_{solvent}$ is negligible i.e

$$\Lambda_m = \frac{K_{solute} \times 1000}{M}$$

Q39 **The solubility of AB if $\lambda^{\circ}_{A^+} = 50\Omega^{-1}cm^2mol^{-1}, \lambda^{\circ}_{B^-} = 70\Omega^{-1}cm^2mole^{-1}$ and the measured resistance was 33.5Ω in a cell with cell constant of $0.2\,cm^{-1}$ is:**

(A) 59.7 millimoles/L **(B) 49.7 millimoles/L**
(C) 39.7 millimoles/L **(D) 29.7 millimoles/L**

Sol. **(B)**

$50 + 70 = \Lambda^{\circ}_m(AB)$ (for sparingly soluble salt)

$$\frac{k \times 1000}{s} = \Lambda_m = \Lambda^{\infty}_m$$

We know that

$$K = \frac{\text{cell constant}}{\text{Resistance}} (C.C.)$$

$$S = \frac{(C.C)}{R} \times \frac{1000}{\Lambda_m} = \frac{0.2}{33.5} \frac{mol}{cm^3}$$

$$= 4.95 \times 10^{-5} \frac{mol}{cm^3}$$

or 49.5 m.moles / liter

Q40 **The conductivity of a saturated solution of Ag_3PO_4 is $9 \times 10^{-6} Sm^{-1}$ and its equivalent conductivity is $1.50 \times 10^{-4} Sm^2$ equivalent^{-1}. The K_{sp} of Ag_3PO_4 is**

(A) 4.32×10^{-18} **(B) 1.8×10^{-9}** **(C) 8.64×10^{-13}** **(D) None of these**

Sol. **(A)**

$K = 9 \times 10^{-6} 65.m^{-1}$

$\Lambda_m = \Lambda_{eq} \times N - fac$

$= \left(1.5 \times 10^{-4}\right) \times 3$

$= 4.5 \times 10^{-4} Sm^2.mol^{-1}$

$$4.5 \times 10^{-4} = \frac{k}{1000 - s}$$

$$s = \frac{9 \times 10^{-c}}{4.5 \times 10^{-1}}$$

$= 2 \times 10^{-5} \text{mol / l}$

$$Ag_3PO_4(s) \rightleftharpoons 3Ag^+(aq) + PO^{3-}_{4(aq)}$$

At equilibrium a-s 3s s

$k_{sp} = 27s^4$

$= 4.32 \times 10^{-18}$

Q41 **We have taken a saturated solution of AgBr K_{sp} of AgBr is 16×10^{-14}. If 10^{-7} mole of $AgNO_3$ are added to 1 litre of this solution find conductivity (specific conductance) if this solution in terms of 10^{-7} S $m^{-1}mol^{-1}$.**

Sol. Given: $\lambda^0_{(Ag^+)} = 6 \times 10^{-3} S\ m^2mol^{-1}$;

$\lambda^0_{(Br^-)} = 8 \times 10^{-3} S\ m^2mol^{-1}$;

$\lambda^0_{(NO_3^-)} = 7 \times 10^{-3} S\ m^2mol^{-1}$;

$(K)_{soln} = (K)_{Ag^+} + (K)_{Br^-} + (K)_{NO_3^-}$

Where $K = \lambda_m \times 1000$

$$AgBr(s) \rightleftharpoons Ag^+_{(eq)} - Br^-_{(eq)}$$

t_{eq} $4 \times 10^{-7}_m$ $4 \times 10^-$

Given: If 10 mole of Ag^+ are added i.e $[Ag^+]_{total} = 5X\ 10^{-7}$

After new $(t_{eq})_2$ $\left(5 \times 10^{-7} - x\right)$ $\left(4 \times 10^{-7} - x\right)$

$16 \times 10^{-14} = \left(5 \times 10^{-7} - x\right)\left(4 \times 10^{-7} - x\right)$

$= 20 \times 10^{-14} - 9 \times 10^{-7}x + x^2$

Solving Quadratic

$x^2 - 9 \times 10^{-7}x + 4 \times 10^{-14} = 0$

$$x = \frac{9 \times 10^{-7} - \sqrt{81 \times 10^{-14} - 16 \times 10^{-14}}}{2} = \frac{10^{-7}}{2} m$$

On solving, new concentration of ions after second equilibrium is achieved

$\left[Ag^{+}\right] = 4.5 \times 10^{-7} M, \left[Br^{-}\right] = 3.5 \times 10^{-7} M, \left[NO_3^{-}\right] = 10^{-7} M$

$K_{sol^n} = 10^{-7} \times 1000\left[4.5 \times 6 + 3.5 \times 8 + 7\right] \times 10^{-3}$

$= 10^{-7} \times \left[27 + 28 + 7\right]$

$= 6.2 \times 10^{-6} S.m^{-1}$

Ionic mobility (μ) :

Defined for ions as follow :

$$\mu = \frac{\text{speed of ion}}{\text{potential gradient}} \qquad \left(\frac{m/s}{v/m}\right)$$

where

$$\text{Potential Gradient} = \frac{\text{Potential difference between electrodes}}{\text{distance between electrodes}} \quad \left(\frac{v}{m}\right)$$

Unit of ionic mobility: $\boxed{\mu \rightarrow m^2.Volt^{-1}.s^{-1}}$

For an ion the ionic mobility is related to its molar conductance (λ_m) as follow:

$$\boxed{\mu_{(ion)} = \frac{\lambda m(ion)}{\text{magnitude of charge} \times F}}$$

Units: $\frac{\Omega^{-1}m^2.mol^{-1}}{C.mol^{-1}}$

$= \frac{mL}{\Omega(amp.S)}$

$\Rightarrow m^2.Volt^{-1}.s^{-1}$

Note:

Ionic mobility is to be used to calculate molar conductivity.

e.g. $\mu_{(Na^+)} = \frac{\lambda m\left(Na^{+}\right)}{1F}$

$\mu^{\infty}_{(Mg^{2+})} = \frac{\lambda^{\infty}_m\left(Mg^{2+}\right)}{2F}$

Conductometric titration :

AIM → Monitoring of the conductance of solution present in conical flask upon addition of solution from burette.

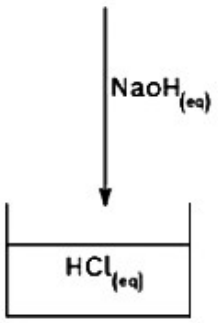
Rules :

(i) For equimolar solution, conductance due to strong electrolyte is more than that of weak electrolyte.

(ii) If type of electrolytes is same then compare conductance by type of ions. (ionic mobility)

(iii) Since H^+ & OH^- ions (lightest cation and anion respectively), have highest ionic mobilities hence they give maximum contribution to the conductance of solution.

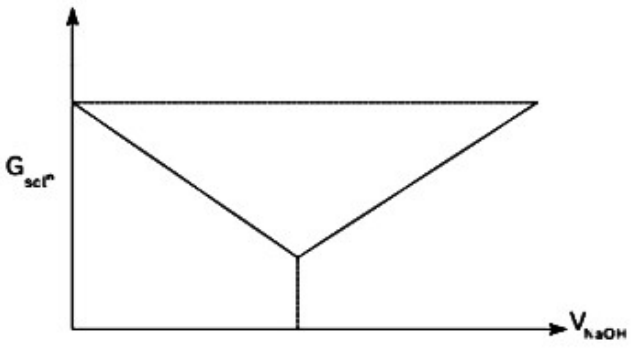
Suppose H^+ replaced by M^+ ion i.e. $[H^+]$ decrease and $[M^+]$ increase, sharp change in conductance is observed due to M^+ being larges in size and having low ionic mobility as compared to H^+.



Titration of $HCl_{(aq)}$ vs $(NaOH)_{(aq)}$ (Titration of strong acid and strong base)

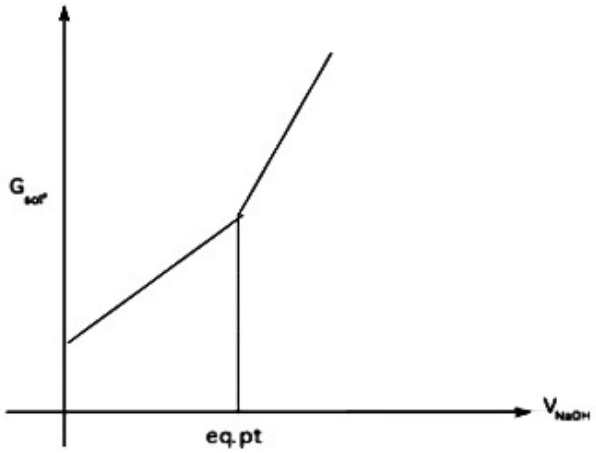
At t = 0	Only $HCl_{(eq)}$ is available in flask ⇒ Strong electrolyte releases H^+ ions with very high conductance
Between, t = 0 to t = equivalence point.	Reaction occurs $HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O(l)$ NaOH added will act as limiting reagent. HCl in flask converted to NaCl. Comparing by cation : H^+ replaced by Na^+ i.e conductance (G) decreases sharply .
At equivalence point	Solution contains only NaCl as electrolyte. On further adding NaOH Concentration of OH^- increases i.e conductance increases sharply.

Hence, the titration curve is:



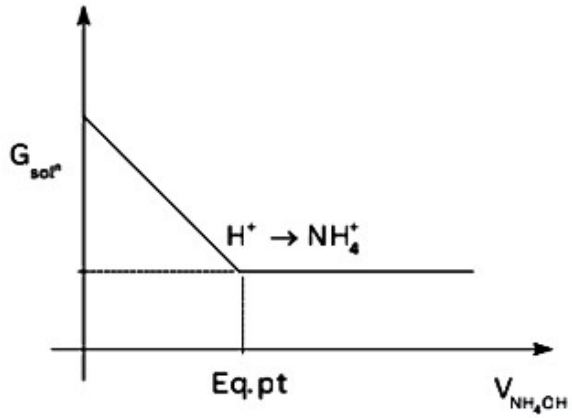
Titration of $CH_3COOH_{(aq)}$ vs $NaOH_{(aq)}$ (Titration of weak acid and strong base)

$V_{NaOH} = 0$	$0 < V_{NaOH} <$ eq.pt.	$V_{NaOH} \rightarrow$ aq.pt.
Only CH_3COOH present $\Rightarrow$ weak acid $(\alpha < 1)$ low $[H^+]$ $\Rightarrow$ low Conductance	Reaction occurs $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$ Throughout this range NaOH acts as limiting reagent. Hence weak electrolyte converted to strong (salt) Weak $\rightarrow$ Strong i.e. G increases slowly	After equivalence point on adding NaOH. No further reaction takes places and $[CH_3COONa]$ remain constant. But, due to addition of NaOH $\rightarrow \left[OH^-\right] \uparrow$ Conductance increases sharply.

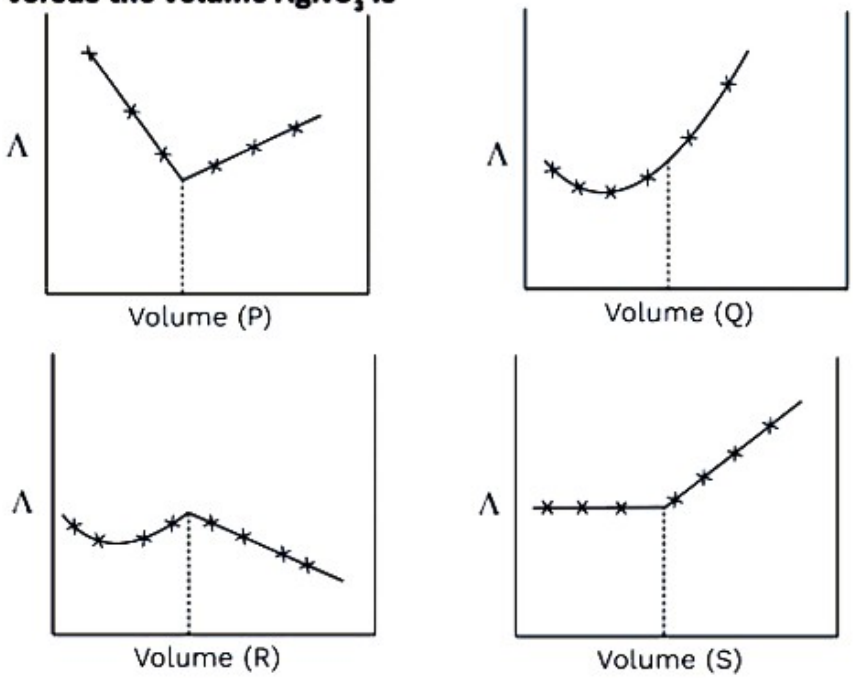


Titration of $HCl_{(aq)}$ vs $NH_4OH_{(aq)}$ (Titration of strong acid and weak base)

At t = 0	Only HCl present i.e., high conductance due to H^+ and Cl^- ions
Between, t = 0 to t = equivalence point.	$HCl + NH_4OH \rightarrow NH_4Cl + H_2O$ Conversion from strong electrolyte $\rightarrow$ Strong electrolyte But due to conversion of cations $H^+ \rightarrow NH_4^+$ G $\downarrow$ Sharply decreases.
After equivalence point	$[NH_4Cl]$ = constant H^+ replaced by NH_4^+ No further reaction takes place after equivalence point. Same cation NH_4^+ remain in solution. NH_4OH being weak base ionizes negligibly, hence change in $[OH^-]$ is negligible consequently, increase in conductance is negligible.



Q42 **$AgNO_3$(aq) was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. The plot of conductance (Λ) versus the volume $AgNO_3$ is**



(A) P **(B) Q** **(C) R** **(D) S**

Sol. **(D)**

Q43 **An aqueous solution of X is added slowly to an aqueous solution of Y as shown in List I. The variation in conductivity of these reactions is given in List II. Match List I with List II and select the correct using the code given below the lists.**

List-I		List-II	
(P)	$\underset{X}{(C_2H_5)_3N} + \underset{Y}{CH_3COOH}$	(1)	Conductivity decreases and then increases
(Q)	$\underset{X}{KI(0.1M)} + \underset{Y}{AgNO_3(0.01M)}$	(2)	Conductivity decreases and then does not change much
(R)	$\underset{X}{CH_3COOH} + \underset{Y}{KOH}$	(3)	Conductivity increases and then does not change much
(S)	$\underset{X}{NaOH} + \underset{Y}{HI}$	(4)	Conductivity does not change much and then increases

Codes:

(A) P Q R S
3 4 2 1

(B) P Q R S
4 3 2 1

(C) P Q R S
2 3 4 1

(D) P Q R S
1 4 3 2

Sol. **(A)**